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About us

The inaugural issue of the NCRB Journal was released in 2022 by National Crime Record Bureau (NCRB). It is an annual English publication that features articles on police related subjects like police science, law, scientific investigation, criminology, forensic science, fingerprint science, physical / biological / behavioral measurements, biometric science, contemporary legal issues, cyber-crime, cyber terrorism, cyber security, socio-economic crime, prisons, etc.

The contributors to its articles include senior police officers and scholars from various prestigious institutions across the country. The articles are reviewed by a Board of Referees before publishing. It is being distributed free of cost to various dignitaries of Ministry of Home Affairs (MHA) and Higher Officers of States/UTs viz., Chief Secretaries, DGPs, Heads of CPOs, SCRB heads, Police Training Colleges etc.



EDITORIAL



It gives me immense pleasure in bringing out the first issue of NCRB Journal. This is a peer-reviewed and multi-disciplinary journal with well-researched articles on topics such as Police Science, Criminology, Forensic Science, Finger Print Science, Contemporary Legal Issues, Cybercrimes, etc. In addition to the articles on new advances in traditional and more established areas of Crime and Policing, the Bureau has always endeavored to include findings on emerging issues.

In the article on **“CyTrain: Cybercrime training Portal with Simulated Learning”**, the authors have given the background, objectives and salient features of CyTrain under 14C which aims to assist State law enforcement agencies in all aspects regarding cybercrime reporting, training, forensics and research.

In the article **“Obfuscation is a growing challenge – A Review”**, the author has highlighted various techniques of alteration used by the criminals to fulfill their ulterior motive and the need to build efficient Fingerprint Identification System to curtail such criminal activities.

In the article **“Security of Women-Technological Issues”**, it is emphasized that the use of the internet, which has an all-encompassing reach, can immensely help in the investigation of cases.

In the article **“Pharmaceutical Crimes: Trends, Patterns, and Challenges in India”**, the author has highlighted the different aspects of pharmaceutical crimes regarding synthetic drugs in every form such as Tablets, Capsules, Powders, Syrups, Injections etc.

In the article **“Every Contact Leaves a Trace: The DNA”**, the authors have highlighted the importance of DNA as a piece of evidence that is a witness in itself.

In the article **“Combating COVID-19 in Correctional Homes of West Bengal”**, the author has highlighted steps taken to combat Covid-19 in prisons such as circulation of guidelines for awareness generations, personal safety measures, social distancing, sanitation & hygiene, medical attention, decongestion, supervision, activities in correctional homes and subsequent measures.

In the article **“Electricity Theft: Overview, issues, prevention and control in India”**, the authors have highlighted issues involved in electricity theft, its prevention, steps being taken by the State Governments in this regard and possible steps for ensuring effective control on electricity theft have also been dealt in the paper.

In the article **“Project Management of CCTNS”**, the authors have highlighted the modernization of the Police to improve outcomes in the area of crime investigation, criminal detection, information gathering and its dissemination across various police organizations and units in the country and to enhance services to citizens.

In the article **“Anticipatory Bail for Children in Conflict with law: Issues & Challenges”**, the authors have highlighted the relevant provisions and the judgments expounding on this issue.

In the last article **“Haryana 112: Emergency Response is just a call away!”** the author has examined the aspect of implementation of Haryana’s 112 project launched to benefit 25 million citizens of the Haryana State.

I thank all the esteemed authors, who have sent their papers for this issue of the Journal. I also thank all the distinguished members of the Board of Referees for their valuable advice. I am sure that readers will find these articles thought-provoking and encourage them to further contemplate on the relevant issues.



(Vivek Gogia)



CyTrain: Cybercrime Training Portal with Simulated Learning

Sanjay Mathur, IPS, Joint Director, NCRB
Dr. Prashun Gupta, Deputy Director, NCRB

Abstract

Capacity building for cybercrime investigation is critical given the rising trend of such crimes. Under the NCTC, NCRB has developed a Massive Open Online Courses (MOOC) platform, namely the CyTrain portal for imparting training to law enforcement agencies at a mass scale on cybercrime investigation-related matters. It offers professional-quality e-learning resources to all law enforcement agencies, prosecutors, judges and other stakeholders. The key feature of this facility is to provide a simulation-based environment for learning about cybercrime investigation.

This paper illustrates the detailed information and architecture of the CyTrain portal, which aims at creating an online learning environment for enhancing the efficiency of cybercrime investigation among field-level officers of the Indian Policing System.

Keywords – CyTrain, NCTC, I4C, Cybercrime, Investigation, NCRB, MOOC, TCSP, MSP, MOODLE.

Introduction

Cybercrime is becoming a worldwide phenomenon and due to its borderless global reach, the traditional law enforcement approach might not be suitable to identify, investigate and prosecute cybercriminals. With the second-largest internet-using population in the world, India also faces a serious problem with cybercrime. Each year, the number of cybercrimes reported across the country continues to increase at a significant pace. The nature of crimes, on the other hand, ranged from petty online frauds to lottery scams and sexual harassment. The sector most targeted though seems to be banking and finance segment. A higher risk extends to other sectors as well with the onset of the coronavirus pandemic and due to most services moving to the online space.

To overcome these challenges, there is an urgent need to sensitize and train the police personnel of all ranks in the latest tools and technology and equip them with expertise to handle digital evidence and investigate cybercrimes. To train such a large number of police personnel through a traditional classroom approach is a huge challenge and will not be able to bolster the pace of capacity-building efforts.

Background

NCRB was set up in 1986 to function as a repository of information on crime and criminals to assist the investigators in linking crime to the perpetrator and empower Indian Police with Information Technology. Ministry of Home Affairs, Govt. of India, has launched the Indian Cyber Crime Coordination Centre (I4C) scheme to act as a nodal point in the fight against cybercrime and assist State law enforcement agencies in all aspects of cyber crime reporting, training, forensics and research. It also facilitates the exchange of information and coordination amongst various law enforcement agencies.

In the current scenario where the number of cybercrimes is increasing, the State Law Enforcement Agencies (LEAs) must have cyber intelligence, investigation and forensic units that are fully equipped with digital forensic tools and expertise. Being an organization set up to empower Indian Police with Information Technology, NCRB, under the aegis of I4C, has been mandated to establish a National Cybercrime Training Centre (NCTC) and develop a Massive Open Online Courses (MOOC) platform to offer professional quality e-learning courses at mass scale in cybercrime investigation related matters to Law enforcement officers, prosecutors, judges and other stakeholders.

Constraints

As such lot of material is already available on the internet in the form of PDFs, Videos, etc. for understanding cybercrime investigation. However, trainees are unable to pay attention as browsing for the correct information required on the internet seems a cumbersome job. The real challenge was to make course content interesting and easy to understand for the police personnel at the ground level so that they are not bogged down by jargon. Therefore, one of the unique requirements was to develop e-learning content for the platform to give a simulated learning experience to the learner. This was not done earlier either in India or elsewhere in the world in the field of cybercrime investigation training. This would mean the creation of scenarios, including digital exhibits (logs, etc.) for extraction by trainees using forensic tools preloaded on the infrastructure, using appropriate procedures.

Opportunities

Training a large number of police personnel across India is not possible without the use of technology for delivering quality training. While NCRB imparts physical training in CCTNS/IT and cybercrime investigation to police officers, the mandate to establish NCTC was seen as a huge opportunity by officers in NCRB to develop new skills in developing MOOC platform and designing simulated e-learning content for cybercrime investigation techniques. A state-of-the-art e-Cyber Lab was also to be established, equipped with the latest digital forensic tools, which are integrated

with the MOOC platform for providing virtual experience on experimentation on the latest modus operandi on cybercrime.

Narrative

NCRB used Massive Open Online Course (MOOC) platform technology to develop infrastructure to facilitate at least 36000 users' concurrency (assuming that approx. 1000 users per State/UT out of 36 States/UTs accessing the training platform simultaneously). NCRB has developed a MOOC platform called the CyTrain portal using the open-source MOODLE framework. Presently, the platform is based on the latest version of MOODLE 3.9 and supports 10,000 user concurrency. The portal supports both English and Hindi languages.

To begin with, NCRB signed MoUs with the Consortium of Educational Communication (CEC) and National e-Governance Division (NeGD)/MeiTY for obtaining e-content for cybercrime investigation in the form of PDFs, PPTs, Videos, etc. The same was uploaded on the CyTrain portal and to date, NCTC has more than 1322 e-learning materials including videos, eBooks, PDF, PowerPoint Presentations, etc. At the same time, a Request for Proposal (RFP) was prepared to select a vendor through open tender to develop professional-quality interactive e-learning content with simulation and gamification activities. A committee under the chairmanship of the Joint Director (CCTNS), NCRB was also formed to validate the course content developed by the vendor. The committee comprises senior police officers from States/UTs, CBI, NPA, and field officers having experience in handling cybercrime investigations. Representatives of NFSU and Judicial officials from High Court Delhi are also members of this committee.

The CyTrain portal has the following six tracks under which course modules have been developed. For each track, the courses have been developed for basic, intermediate and advanced levels:

- a) Responder Track (For First Responder and Duty Officer)
- b) Forensic Track (For Digital Forensics Specialist)
- c) Investigation Track (For general Investigator and Cybercrime Investigator)
- d) Intelligence Track (For Cybercrime Intelligence Officers /Analysts)
- e) Management Track (For Cybercrime /Digital Forensics Head of Unit and Heads of Police Forces)
- f) Judiciary / Prosecution Track (For Judges / Prosecutors)

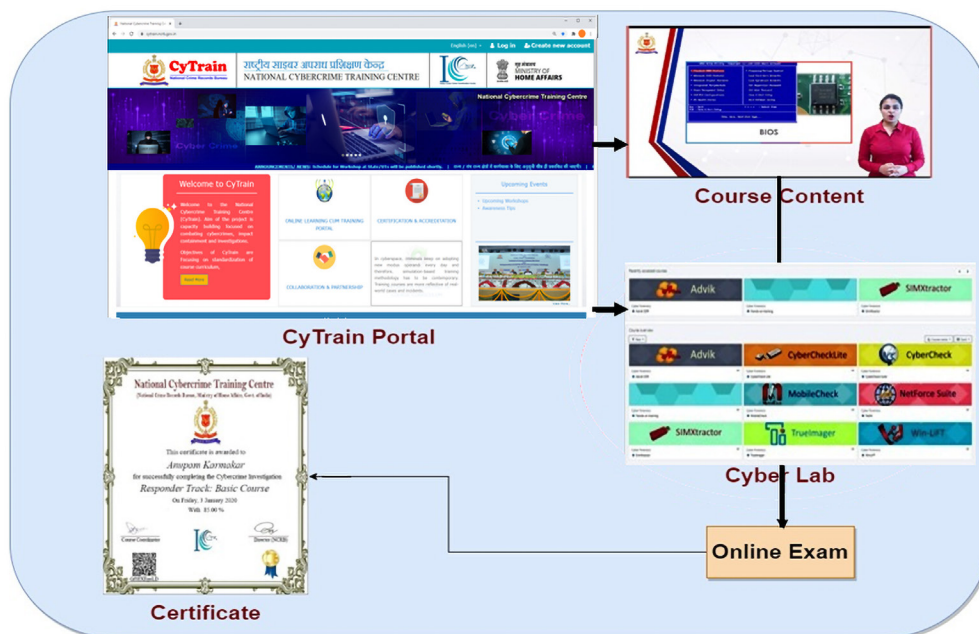
CyTrain Dashboard has two types of course display:

- a. Recently Accessed Courses: Courses that are recently accessed by the user.
- b. Course Overview: Courses in which the user is enrolled.

User can register himself/herself in any online course with their Government email ID and undertake the course at their own convenient place. Users can access a course by just clicking on the Course icon or course name, after which the user will see the following screen:



At the end of the training, he has to undergo an online test, passing which will entitle him to a Certificate. The workflow of the CyTrain portal is depicted as below:

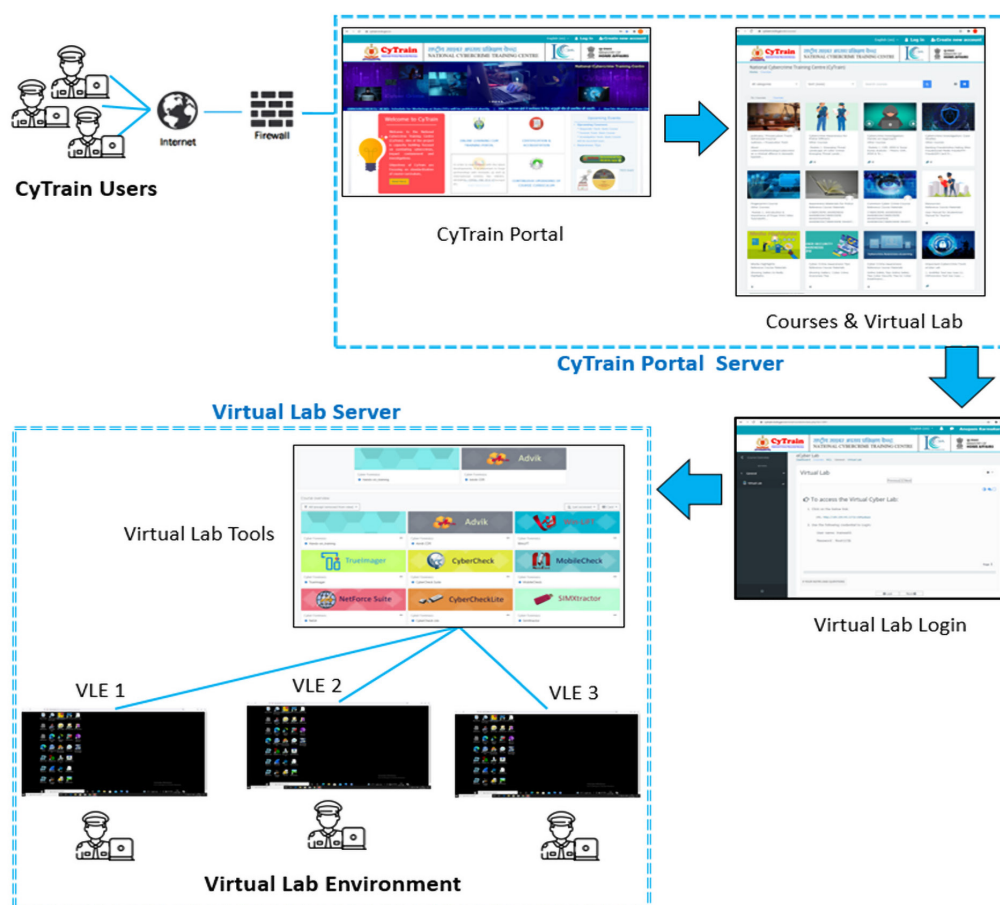


Certificate generated through CyTrain portal contains QR code so that in case of any doubts, it can be verified on this portal. A sample certificate is also depicted below:



e-Cyber Lab

NCRB has also established a State of Art e-Cyber Lab through C-DAC at NCRB Hqrs, New Delhi. The Lab is equipped with more than 25 latest Digital Forensics tools from the industry and more than 50 open-source cybercrime detection and forensics tools. The e-Cyber Lab is also integrated with the MOOC platform (CyTrain portal) and is called as e-Cyber Lab which was launched on 13.10.2020 for providing a virtual experience of experimentation on the latest modus operandi on cybercrime. The officers can log into e-Cyber Lab remotely and learn about and use any tool on pre-arranged use cases or by importing their problem statement. This provides access to these sophisticated tools to all trainees to get real hands-on experience through this virtual lab. The basic architecture of eCyber Lab is depicted as below:



Continuous up-gradation of content

To continuously upgrade the MOOC platform-based CyTrain portal, a team of MOOC experts is deployed to maintain and upgrade the portal to enhance its features and capabilities. The e-content consists of presentations, simulation-based contents and quizzes along with the latest case studies. The course content is continuously reviewed and upgraded at regular intervals to equip the trainees with knowledge of the latest tools and expertise to handle digital evidence and investigate cybercrimes and upgrade them about the latest trends and modus operandi by cybercriminals.

The design and development of course content were planned to be completed within a year, however, the ongoing Covid-19 pandemic has been a major disruptor and posed a serious challenge in achieving this target. As of now, more than 50% of the task has already been completed.

To overcome the challenges posed by the covid-19 pandemic, extensive online training programs are planned and executed by NCRB to sensitize the police personnel of all ranks about the CyTrain portal and courses available on it as well as the latest cybercrime case studies. For this purpose, a team of experts is hired to work as 'Training Co-ordination Service Provider' (TCSP) to conduct online training programs. It has been a very successful initiative by this Bureau. Till date, 68 half-day online live training programs have been conducted where 9500 participants of all ranks from various States/UTs/Police Training Institutes have attended these courses on CyTrain Portal where latest cybercrime case studies were also presented. Some snapshots of live online training sessions are provided below:



Current Usage

As on November 2022, the total number of trainees registered on the CyTrain portal for self-paced sources is around 27800 and around 7200 certificates have been issued to those who have completed the courses. During various online training sessions with State Police, it is understood that most of the Police Officers/Officials at the field level do not know about NCTC/CyTrain portal. Some of the Officers/Officials who knew about CyTrain but do not have govt. domain email id (which is a mandatory requirement) for self-registration on this portal. Alternatively, for offline registration, most of them were also not able to forward their details (with private email id) through State CCTNS nodal officers.

Popularization of Portal and Training Workshops

To popularize the CyTrain portal among all levels of Police Officers/ Officials, a need was felt for organizing the 'Sensitization Campaign' of the CyTrain portal across all the States/UTs by demonstrating various features and highlighting its course content. These workshops would also include live lecture/ hands-on sessions from experts on cybercrime investigation. The objective of these workshops is to produce a good number of registrations on the CyTrain portal in a short period.

To organize these workshops in States/UTs, NCRB on boarded a Training Co-ordination Service Provider (TCSP). The TCSP includes the resource manpower like capacity building expert cum course coordinator, Cyber forensic experts and technical assistants for imparting live lectures and hands-on training on cybercrime investigations. TCSP by itself under the guidance of NCRB has planned and designed courses, apart from that , it also co-ordinates with States/UTs for getting the nomination of trainees, arranges and assigns instructors for live lectures/hands-on, assessment of trainees, issuance of certificates, trainee's feedback analysis and overall supervision as well as budgetary requirements of conducting courses etc.

As part of a pilot run, seven such workshops were conducted on online mode and it was observed that more than 2500 trainees participated from different States and UTs.

NCRB has also printed a poster as shown below on CyTrain and distributed it to all the police stations in the country with a request to prominently display it to popularize the portal among all ranks of police officers.



CyTrain
National Crime Records Bureau

राष्ट्रीय साइबर अपराध प्रशिक्षण केन्द्र
NATIONAL CYBERCRIME TRAINING CENTRE





गृह मंत्रालय
MINISTRY OF HOME AFFAIRS

NATIONAL CYBERCRIME TRAINING CENTRE (CYTRAIN)



**VIRTUAL TRAINING CENTRE | MASSIVE OPEN ONLINE COURSES (MOOC) PLATFORM
| TRAINING BASED ON SIMULATED ENVIRONMENTS |**

WHO CAN ACCESS THE PORTAL*?	HOW TO ACCESS THE CYTRAIN?	WORKFLOW OF CYTRAIN!
👉 STATE/UT POLICE 👉 CPO/CAPF 👉 FORENSIC DEPARTMENT 👉 JUDICIARY/PROSECUTION 👉 ANY GOVT. OFFICER WHO HAS VALID GOVT EMAIL ID	OPEN URL HTTPS://CYTRAIN.NCRB.GOV.IN/ AND REGISTER YOURSELF BY CLICKING ON "CREATE NEW ACCOUNT"	👤 LOGIN TO CYTRAIN 📖 ACCESS TRAINING COURSES 💡 PRACTICE IN E-CYBER LAB 🏆 ACCREDITATION & CERTIFICATION

***ALL STAKE HOLDERS WHO ARE PART OF CYBER CRIME INVESTIGATION ECO-SYSTEM WITH VALID GOVT EMAIL ID CAN ACCESS THE PORTAL**

BENEFITS

- ✔️ ACCESS TO PORTAL ANYTIME, ANYWHERE AND FREE OF COST
- ✔️ ONLINE INTERACTIVE LEARNING THROUGH SIMULATION & GAMIFICATION
- ✔️ SEAMLESS ACCESS TO LATEST CYBER FORENSIC TOOLS
- ✔️ CERTIFICATE ON SUCCESSFUL COMPLETION OF TRAINING FOR EACH TRACK
- ✔️ RECOGNITION IN NATIONAL & INTERNATIONAL CRIMINAL JUSTICE SYSTEM



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Learning / reflection:

The keen focus of NCTC is to provide world-class training on Cybercrime related courses for all Officers dealing with Cybercrime investigations across the country. More than 11800 users are currently registered in the CyTrain portal and around 2930 certificates have been issued to those who have completed the courses. NCTC is the first initiative in the country to provide massive open-source online training for cybercrime-related investigations. Further, a virtual Cyber lab that is associated with the CyTrain portal is the best add-on for hands-on experience. A team of experts associated with NCTC regularly updates the course contents as per the latest trends in cybercrimes. The main objective of NCTC is capacity building and to create a team of experts in every unit across the country in the field of cybercrime Investigation who can effectively deal with cybercrimes in the country.



Obfuscation is a Growing Challenge – A Review

Ravinder Kumar, Director, CFPB, NCRB
Movitha Mohandas, SI (FP), CFPB, NCRB

Introduction:

Fingerprint is an amazing science enacted by the fascinating loops, whorls and arches etched on the fingers of a human being. From earlier times, criminal identification by means of fingerprints is one of the most potent factors in obtaining the apprehension of fugitives because of its peculiar adaptability to the field. This type of identification also makes possible an accurate determination of the number of previous arrests and convictions which of course, results in the imposition of more equitable sentences by the judiciary. Recent advances in automated fingerprint identification technology, coupled with the pronounced need for reliable person identification, have resulted in the increased use of fingerprints in both governmental and civilian applications such as border control, employment background checks and secure facility access.

The numerous fingerprint identification systems with at most accuracy in identifying individuals have prompted the engagement of individuals in extreme measures for the purpose of circumventing the systems. The deliberate attempt by an individual to mask his/her identity from a biometric system by altering the biometric trait before its acquisition by the system is known as obfuscation. By the subjective assessment of altered fingerprints, the alteration can be divided into three types. They are obliteration, distortion, and imitation. Obliteration is performed by cutting, abrading, burning and poring strong chemicals into ridge patterns. Skin diseases also obliterate the fingerprint. Obliteration is further divided into scar and mutilation. Fingerprint can be distorted by changing the ridge structure by plastic surgery in which portions of the skin are removed from the finger and grafted back in different positions. Imitation type of alteration is obtained by performing a surgical procedure in such a way that the altered region looks like natural fingerprint. These surgeries may involve the transplantation of a large area of skin from other parts of the body such as fingers, palms, toes, and soles. It is very difficult to detect imitation type of alteration because of its similarity with natural fingerprints.

The primary purpose of fingerprint alteration is to evade identification using techniques varying from abrading, cutting and burning fingers to performing plastic surgery. The use of altered fingerprints to mask one's identity constitutes a serious "attack" against a border control biometric systems since it defeats the very purpose for which the system has been deployed in the first place, i.e., to identify individuals in a watch-list. It should be noted that altered fingerprints are different

from fake fingerprints. The use of fake fingers - made of glue, latex, or silicone - is a well-publicized method to circumvent fingerprint systems. Altered fingerprints, on the other hand, are real fingers that are used to conceal one's identity to evade identification by a biometric system. Thus, fake fingers are typically used by individuals to adopt another person's identity while altered fingers are used to mask one's own identity.

This article is an attempt to consolidate various studies on obfuscation or fingerprint alteration by criminals as well as civilians to suit their specific purpose of blinding the advanced fingerprint identification systems.

Case studies:

Numerous cases of fingerprint alteration have been reported till date. Some of the high profile cases include:

- An infamous bank robber and murderer, "Gus" Winkler altered his fingerprints by tearing the flesh of his fingers except for the thumb. It was found that the whorl pattern in one of his finger even changed to loop. This case was reported in 1933.



Fig.1 Fingerprints of "Gus" Winkler before and after alteration (whorl to loop)

- In 1934, a notorious criminal and bank robber, John Dillinger applied acid to his fingertips to destroy fingerprints. But still, his fingers had sufficient information to identify as he had only burned a small area of his fingertips.

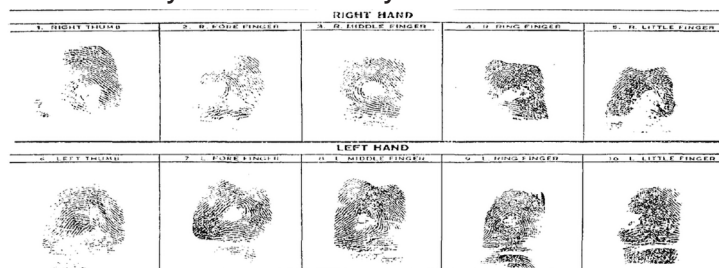


Fig 2. Fingerprints of John Dillinger

- In 1941, a habitual offender, Roscoe Pitts (also known as Robert J Philipps) hired a plastic surgeon to remove the skin of his fingertips and get it replaced by skin grafted from the chest. But his true identity was revealed to his surgeon after his arrested and confirmed by comparing the second joints of his fingers with his fingerprint card.
- In 1995, Alexander Guzman was arrested for possession of a false passport by Florida officials. He was found to possess mutilated fingerprints. After an extensive search, the reconstructed fingerprints were linked to the fingerprints of an absconding drug dealer Jose Izquiere. He had carried out a three-step mutilation including making a 'Z' shaped cut, lifting and switching two triangles, and then stitching together.

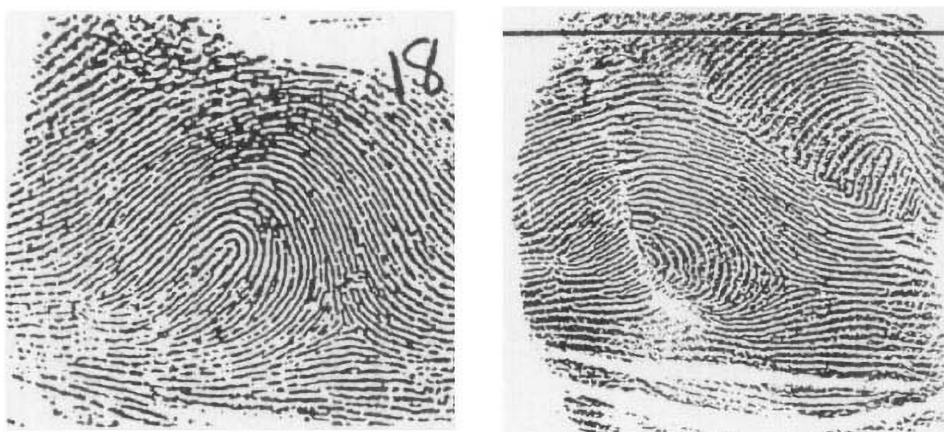


Fig 3. Fingerprints of Jose Izquiere before and after mutilation.

- In 2005, a drug dealer Marc George got his fingerprints replaced with the skin from his feet with the help of a plastic surgeon.
- In 2008, Edgardo Tirado was arrested on drug charges and was found to have altered his fingerprints by cutting his fingertips and stitching to avoid being linked to his criminal antecedents.

The alteration of fingerprints is not only used by criminals but also by several civilians to get immigration clearances and cross borders.

- In 2008 a woman deceived two thumb-based Taiwan border control identification systems by performing plastic surgeries to alter both her face and fingerprints.
- A lady named Lin Rong was arrested in Japan. The International Crime Office of Japan's National Police Agency reported that she had got the prints of left and right hands swapped to illegally enter Japan. The officials of this same office have also reported that have caught eight other people who tried to alter their fingerprints through various techniques.

- It was also reported in Sweden and France that hundreds of asylum seekers had cut, abraded and burned their fingertips to prevent identification by EURODAC, a European Union-wide fingerprint identification system for identification of asylum seekers.

Apart from these intentional alterations recent studies from the Netherlands have reported that chemotherapy agent capecitabine (Xeloda) can cause fingerprint loss. Researchers from the Erasmus MC Cancer Institute said that hand-foot syndrome is a common side effect of treatment with capecitabine in 50-60% of patients. Mr. Ron Mathijssen, a medical oncologist and author of the study stated in an interview that the patients should be warned of the loss of fingerprints as it may cause trouble to their day-to-day lives, especially if traveling to a foreign country. In 2009, a 62-year-old man traveling from Singapore to U.S. was detained because of fingerprint loss. He was a cancer patient who had been treated with capecitabine since 2005. The man was eventually released after he was found not a threat. Other issues may also arise at home, such as access to personal electronic devices that often use fingerprint identification like smartphones and laptops.

Conclusion:

The increased number of reported cases of fingerprint alterations also cautions us about the number of individuals entering into fingerprint alterations. This has attracted the attention of researchers towards the development of more efficient algorithms which could detect altered fingerprints. Different methods have been developed for fingerprint alteration detection by the analysis of ridge orientation field, minutiae distribution, reliability of ridge orientation, and density of singular points.

The studies of Vidya and Phulpagar, 2017 has shown that the Altered Fingerprint analysis SVM classifier could be trained to classify altered and normal fingerprints using registered ridge orientation and period map. Similarly, many studies have been reported on algorithms to study distortions and scars also. Identifying fingerprint alteration using the reliability map of the orientation field was reported by Adina Petrovici and Corneliu Lazar in the paper presented in Automatic Control and Applied Informatics Department, Technical University "Gheorghe Asachi", Iasi, Romania.

Still more efficient algorithms need to be designed and implemented with Automated Fingerprint Identification System (AFIS) to effectively check criminals using fingerprint alterations to curtain their identity and retrieval of previous criminal records. Technology should be expanded to counter these attempts to ensure that law enforcement agencies across the world are ahead of criminals.

References:

1. J. Feng, A. K. Jain, A. Ross, "Fingerprint Alteration", MSU Technical Report, MSU-CSE-09-30, Dec. 2009.
2. Anoop.T.R, Deepak.P, Jeevan K.M," Altered Fingerprint Detection Using Scar Feature" IOSR Journal of Electronics and Communication Engineering, 2016.
3. H. Cummins, "Attempts to Alter and Obliterate Finger-prints," Journal of American Institute of Criminal Law and Criminology, vol. 25, pp.982–991, 1935.
4. History of Fingerprint Removal, <http://jimfisher.edinboro.edu/forensics/fire/print.html> .
5. K. Wertheim, "An Extreme Case of Fingerprint Mutilation," Journal of Forensic Identification, vol. 48, no. 4, pp. 466–477, 1998.
6. Woman Alters Fingerprints to Deceive Taiwan Immigration Fingerprint Identification System, October 2008, <http://www.zaobao.com/special/newspapers/2008/10/hongkong/081002r.shtml> (In Chinese)
7. Asylum Seekers Torch Skin off Their Fingertips So They Can't Be ID'd by Police, 2008, <http://www.mirror.co.uk/sundaymirror/2008/06/29/asylum-seekers-torch-skin-off-their-fingertips-sothey-can-t-be-id-d-by-police-98487-20624559/>.
8. Vidya Ramesh Patil and Prof. (Dr). B. D. Phulpagar, Efficient Altered Fingerprint Analysis and Rectification of Distorted Fingerprint, International Research Journal of Engineering and Technology (IRJET), Volume: 04, Issue: 07, July -2017



Security of Women-Technological Issues

- Media and social media-impact on women's security.

- Security Applications

- Cyber Crime and forms of violence against women

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Abstract:

Information technology has widened itself over the last two and half decades and has become the axis of today's global and technical development. The world of internet provides every user all the required information fastest communication and sharing tool making it the most valuable source of information. With the numerous advancement of internet, the crime using internet has also widened its roots in all directions. Cybercrime poses a great threat to individuals. Cyber-crime is a global phenomenon and women are the soft targets of this new form of crime. Cyber-crime is emerging as a challenge for national and economic security. Various issues that are discussed here are - Cyber Stalking, Harassment via E-mail, Cyber Defamation, Morphing, and Email spoofing against Women.

KEY WORDS: Information Technology, cybercrime, E-Mail, Face Book, Twitter, You Tube, what's app

I. Introduction:

Criminal activities that are traditional in nature, such as theft, fraud, forgery, defamation and mischief, all of which are subject to the Indian Penal Code. The abuse of computers has also given birth to a gamut of new age crimes that are addressed by the Information Technology Act, 2000 which has been amended several times and the latest in 2008 has tried to bring fast growing cybercrimes under its ambit.

We can generally categorize cybercrimes in two ways:

1. The computer as a target: - using a computer to attack other computers.

E.g.: Hacking, virus/worm attacks, Dos attacks etc.

2. The computer as a weapon: -Using a computer to commit real world crimes.

E.g.: Cyber Terrorism, IPR Violations, Credit card frauds, EFT Frauds, Pornography, Social media (Face book, Twitter, you tube, Blogs, Pinterest, Instagram, what 'Sapp, etc.)

II. Cyber Laws and Statistics:

Cyber Laws:

#1.Cybercrimes under the IT Act:

→Tampering with computer source documents-Sec.65.

→Hacking with computer systems, data alteration-Sec 66.

→Publishing obscene information-Sec 67.

→un-authorized access to protected system-Sec.70

→Breach of confidentiality and privacy-Sec.72

→Publishing false digital signature certificates-Sec.73

#2.Cybercrimes under IPC and Special Laws:

→Sending threatening messages by E-mail-Sec 503 IPC

→Sending Defamatory messages by E-mail-Sec 499 IPC

→Forgery Electronic records-Sec 463 IPC

→Bogus Websites, Cyber frauds-Sec 420 IPC

→E mail spoofing-Sec 463 IPC

→Web jacking-Sec 383 IPC

→E-mail abuse-Sec 500 IPC.

The national commission for women receiving the complaints In the year 2020-2021:

Sl. No	Category	No. of Complaints
1	Right to live with Dignity	7715
2	Police apathy against women	1276
3	Dowry harassment to married women	3788
4	Violence against women	963
5	Outraging modesty of women	1679

6	Sexual harassments	577
7	Right to exercise choice in marriage	460
8	Cybercrimes against women	704
9	Free legal aid for women	56
10	Privacy of women	115

1. Media and social media-impact on women's security:

The present time is reflective of social media playing an instrumental role in changing the mindset of the people in various ways. It has been a major force behind many social changes, debates, mass movements and creating activists' groups. Everyone is familiar with some of the most brutal and heinous crimes related to rape like Nirbhaya gang rape, Delhi uber rape case, Mumbai shakti mill rape case, etc.

These cases of women atrocities would not have been able to draw such immense attention of the masses without the active employment of social media tools by them and the people expressing their strong opinion on the different facets of the case. Times are such that important happenings and issues become a matter of concern and wide spread debate within seconds of the outbreak of the event with the help of internet.

2. Role of social media platforms:

- 2.1 Create ways for women and representatives of minority groups to highlight reports of harassment, particularly incidents involving multiple accounts or lasting several days, indications that the activity is organized.
- 2.2 Employ local staff, particularly grievance officers, to ensure that workers can adequately evaluate complaints about posts uploaded in local languages based on local cultural context.
- 2.3 Facebook should revisit its real-name policy and make it more flexible, especially for marginalized sections, journalists, and whistle-blowers under threat. Right to privacy and anonymity are fundamental rights and Facebook must uphold them. Harassment continues despite the requirement, yet victims fear it can be used against them.

3. Role of government:

- 3.1. Encourage women to report when criminals violate their rights online
- 3.2 Ensure that the portal for cybercrime prevention against women and

children is accessible via mobile internet as an app that functions even where the band width is low. Ensure that user's privacy is protected and that they are not subject to monitoring.

- 3.3 Implement the law against individuals responsible for inciting and carrying out online violence against women and marginalized communities, regardless of their political or religious agenda.

4. Role of law enforcement agencies:

- 4.1. Foster an environment in which individuals feel confident enough to report online abuse to authorities, even if this challenges cultural norms and takes time.
- 4.2 Stop dismissing reports of sexualized or gender-biased abuse just because they take place on the internet.
- 4.3 Sensitize officers that the response to online harassment is not to stop the victim using the internet.
- 4.4 Educate officers about the laws that apply to online harassment, and how to direct complainants to appropriate legal recourse.

5. Case Laws:

5.1 Ritu Kohli Case:

The perfectly normal married life of Ritu Kohli, New Delhi turned upside down, when she started receiving a number of e-mails from an unknown source. Initially she ignored the mails.

Stalker used obscene and obnoxious language and posted her residence telephone number and other personal details on various websites, inviting people to chat with her on the phone. As a result, she started receiving numerous obscene calls at odd hours from everywhere. Then she got alarmed. Distraught, Kohli lodged a police complaint. Fortunately, Delhi Police immediately sprang into action. They traced down the I.P Address (Internet Protocol address) of the hacker to a cyber café.

The cyber stalker-Manish Kathuria, later got arrested by the Delhi Police and was booked under Sec.509 of the IPC(Indian Penal Code) for outraging the modesty of a woman and also under the IT Act(Information Technology Act) of 2000.

The Case highlighted here is the **first case of Cyber Stalking** to be reported in India.

5.2 Neha Ghai Case:

A 28-year-old woman, Neha Ghai was shocked after she received objectionable calls and text messages on her mobile phones and even vulgar e-mails in her in box. When she approached the cyber cell and lodged a complaint against the accused, she came to know that she had become a victim of cyber stalking and the stalker had collected all her personal details and posted on objectionable portals.

Cyber stalking nowadays has become a serious problem and victims should immediately inform the police. The police can trace them with their IP.

6. Security apps:

No one has got time to unlock the phone, punch in the number and speak to one's kin in a situation. The phone is probably snatched away or you may not know your location to explain. Some of the useful apps details are as given below:-

- (i) **SOS-Stay safe:** No need to type in the phone unlock code. Just install the app in the android phones and by shaking it a couple of times, the trigger will be alerted along with your geographical location to the emergency contact. It also features home button tapping, which will also send a message, location and even battery status of phone and an audio recording.
- (ii) **Nirbhaya:** Nirbhaya is another safety application for women. Can be easily activated just by pressing the button once. you can customize the activation and alert, such as by shaking, pressing button and sending alert by message or call. This one requires data plan and GPS functioning. It also sends the user location after every 2 hours at a change of 300 meters.
- (iii) **Smart 24/7:** This is the security app linked with the police. The panic button allows you to place a call to the police. Alert calls can also be initiated to the pre-designated emergency contact. If GPRS is not functional, the location is sent via SMS. The app clicks photos and records audio-video of the situation. Has plenty of other features such as tracking, customer care and chat.
- (iv) **Cybercrime cell Help-line,** Visakhapatnam-94906 17917.

7.Cyber-crime and new form of violence against women:

“Cybercrime is a new area and many women are falling prey out of ignorance. Widespread awareness on cybercrime should be taken up by various organizations and establishments at different levels,”-Justice Rohini on 12.03.2017, Visakhapatnam.

Cybercrime against women can be characterized under following heads:-

Amongst the various cybercrimes committed against individuals and society at large the crimes which can be mentioned as specially targeting women are as follows:-

7.1. Harassment via e-mails:

This kind of harassment is not a new phenomenon. Women are victim of e-mail harassment from the very beginning.

Email and IRC related crimes

a) E-mail Spamming: Email “spamming” refers to sending email to thousands and thousands of users-similar to a chain letters. Malicious codes are sent through email. Emails are used to send viruses, trojans etc through emails as an attachment or by sending a link of website which on visiting downloads malicious code.

b) Email bombing:

Email ‘bombing’ is characterized by abusers repeatedly sending an identical email message to a particular address.

c) Sending threatening emails

d) Defamatory emails

e) Email Frauds

7.2. Defamation:

Defamation can be understood as the intentional infringement of another person’s right to dignity. Cyber defamation occurs when defamation takes place with the help of computers and on the internet.

E.g.: Someone publishes defamatory matter about someone on a website or sends e-mails containing defamatory information to all the person’s friends. Information posted to a bulletin board can place defamatory matter against you. Cyber defamation is also called cyber smearing.

The very first instance of cyber defamation in India was recorded in the case of ***SMC PNEMATICS (INDIA) PVT.LT VS. JOGESH KWATRA***

Cyber defamation was reported when a company's employee (defendant) started sending derogatory, defamatory and obscene e-mails about its Managing Director. The E-mails were anonymous and frequent, and were sent to many of their business associates to tarnish the image and goodwill of the plaintiff company. The plaintiff was able to identify the defendant with the help of a private computer expert and moved to the Delhi High court. The court granted an ad-interim injunction and restrained the employee from sending, publishing and transmitting e-mails, which were defamatory or derogatory to the plaintiffs.

Another famous case involving a woman was ***THE STATE OF TAMIL NADU VS. SUHAS KATTI***

The case is related to posting of obscene, defamatory and annoying message about a divorcee woman in the yahoo message group-mails were also forwarded to the victim for information by the accused through a false email account opened by him in the name of the victim. The posting of the message resulted in annoying phone calls to the lady in the belief that she was soliciting.

7.3. Cyber Stalking:

Cyber stalking involves following a person's movements across the internet by posting messages (sometimes threatening) on the bulletin boards frequented by the victim, entering the chat-rooms frequented by the victim, constantly bombarding the victim with emails etc. In general, the harasser intends to cause emotional distress and has no legitimate purpose to his communications.

Paedophiles:

These are persons who intentionally prey upon children. They earn the teenagers' trust and gradually seduce them into sexual or indecent acts. Paedophiles lure the children by distributing pornographic material, then they try to meet them for sex or to take their nude photographs including their engagement in sexual positions.

7.4. Identity theft

Identity theft is the fastest growing crime in countries like America. Identity theft occurs when someone appropriates another person's personal information without his knowledge to commit theft or fraud. Identity theft is a vehicle for perpetrating other types of fraud schemes.

7.5. Pornography

The literal meaning of the term 'pornography' is "describing or showing sexual acts in order to cause sexual excitement through books, films, etc."

This would include pornographic websites, pornographic material produced using computers and use of internet to download and transmit pornographic videos, pictures, photos, writings etc.

Adult entertainment is the largest industry on internet. There are more than 420 million individual pornographic web pages today. Research shows that 50% of the web-sites containing potentially illegal contents relating to child abuse were 'pay-per-view'. This indicates that abusive images of children over internet have been highly commercialized. Pornography delivered over mobile phones is now a burgeoning business, "driven by the increase in sophisticated services that deliver video clips and streaming video, in addition to text and images."

Effects of Pornography

Research has shown that pornography and its messages are involved in shaping attitudes and encouraging behavior that can harm individual users and their families. Pornography is often viewed in secret, which creates rift within marriages that can lead to divorce in some cases.

The DPS MMS Scandal is a very famous case of cyber pornography where a MMS clip of a school girl in compromising situation was made and distributed amongst various internet networks.

In another incident at Mumbai, a Swiss couple gathered slum children and then forced them to appear for obscene photographs, then uploaded those photographs to websites especially designed for paedophiles. The Mumbai police arrested the couple for pornography.

The most recent example is of Delhi Metro CCTV footage leaks case, where the CCTV recording of couples getting intimate in metro stations etc. which was recorded by police security cameras had been leaked on internet.

In addition, pornography promotes the allure of adultery, prostitution and unreal expectations that can result in dangerous promiscuous behavior.

7.6 Privacy

Privacy refers to the right of an individual/s to determine when, how and to what extent his or her personal data will be shared with others. Breach of privacy means unauthorized use or distribution or disclosure of personal information like medical records, sexual preferences, financial status etc.

7.7 Confidentiality

It means non-disclosure of information to unauthorized or unwanted persons. In addition to personal information some other type of information which is useful for business and leakage of such information to other persons may cause damage to business or person. Such information should be protected.

Generally, for protecting secrecy of such information, parties while sharing information form agreement about the procedure of handling of information and to not disclose such information to third parties or use it in such a way that it will be disclosed to third parties.

Many times party or their employees leak such valuable information for monetary gains and causes breach of contract of confidentiality. Special techniques such as social engineering are commonly used to obtain confidential information.

7.8 Morphing

Morphing is editing the original picture by unauthorized user or fake identity. It was identified that females' pictures were downloaded by fake users and again re-posted/uploaded on different websites by creating fake profiles after editing it. This amounts to violation of I.T. Act, 2000 and attracts sec 43 and 66 of the said Act. The violator can also be booked under IPC also. The Times of India reported that Delhi based beautician told the police that her photograph was flashed on a porno portal along with her mobile number.

CASE LAWS

7.9 Air Force Bal Bharati School case (Delhi) -

Where a student of the school was teased by all his classmates for having a pockmarked face. He decided to get back at his tormentors and scanned photograph of his classmates and teachers, morphed them with nude photographs and put them on a website that he uploaded on to a free web hosting service. The father of one of the class girls featured on the website came to know about this and lodged a complaint with the police

7.10 Email spoofing:

A spoofed e-mail is one which misrepresents its origin. It shows its origin to be different from one it actually originates from. A review in the cyberlawtimes.com shows that India has crossed the danger mark in cybercrime targeting women. Child statistics law enforce us confirm, that the maximum number of cyber crimes related to obscenity occurred in Mumbai last year.

The most popular case of cyber spoofing is Gujarat Ambuja's Executive case. In this case the perpetrator pretended to be a girl for cheating and blackmailing the Abudhabi based NRI.

Bangalore, Chennai, Hyderabad, Pune reported only a handful obscene crimes but saw a greater incidence of hacking.

The more common method used by men is to email vulgar photographs of themselves to woman, praising their beauty, and asking them for a date or inquiring how much they charge for 'services', besides sending explicit messages via e-mail, SMS, and chat, many also use morphography-placing the victim's face on another, usually nude, body. In other instance, a couple entered an internet chat room agreeing to strip for each other. The guy stripped, but the person at the other end was actually another man and his friends. They recorded it and uploaded the clip on a promo website. These things happen in every city but only in every 500 cases is reported.

7.11 Black Mailing:

Debarti Halder, Advocate, Madras High court, and an independent counselor for cyber crime victims explains, "the most common felony is using the victims picture or name to defame her. Almost every women victim would have similar tales of harassment. The harasser creates

fake, malicious profiles with the victims picture and invites others to view it, or ask for some favor from the victim and threaten to make such pictures public otherwise. In most cases this turns out to be a revenge taking act.” Often there is this wrong notion that cyber crime is alien to e-world in India. Even with security agencies, such as McAfee keeping a tab on cyber predators, the number of cases in Asia has only spiraled. According to a McAfee report, there has been 150% growth of Trojans (Programs that contain malicious code which when triggered cause loss or theft of data) using social engineering since 2006.

“Just as every aspect of communication has been changed by the internet, crime has evolved to profit from the millions of potential victims connected to one global network. online trading sites for identities creates a market for thieves to sell to more sophisticated criminals,”-McAfee.

8. Provisions of the IT ACT 2000 relating to cybercrime and offences against women;

Chapter xi of the IT Act deals with the offences such as tampering with computer source documents (Sec.65), Hacking with computer system (Sec.66), Publishing of information which is obscene in electronic form.(Sec.67) Access to proceed system (Sec.70), Breach of confidentiality and privacy (Sec.72), Publication for fraudulent purpose (Sec.74) IT Act 2000 still needs to be modified. It does not mention any crime specifically as against women and children.

The elementary problems, which are associated with cyber-crimes, are jurisdiction, loss of evidence and cyber savvy judges are the need of the day. Today with the growing arms of cyberspace the territorial boundaries seem to vanish. Thus the concept of territorial jurisdiction as envisaged under Sec.16 of C.P.C and Sec.2 of the IPC will have to give way to alternative method of dispute resolution.

Again, under no section in IT Act 2008, obscenity for personal viewing is an offence. In fact as in IPC 292 again, if it's proved that you have published or transmitted or caused to be published in the electronic form only then under section 67 IT Act it can be an offence. Last but not the least, the IT Act 2008 does not mention the typical cybercrimes like stalking, morphing and email spoofing as offences.

The rapid increase in the use of computer and internet has given rise to new forms of crimes like, sending offensive emails and multimedia messages, child pornography, cyber terrorism, publishing sexually explicit materials in electronic form, video voyeurism, breach of confidentiality and leakage of data by intermediary-commerce frauds like cheating by personation-commonly known as phishing, identity theft, frauds on online auction sites etc. So, Penal provisions were required to be included in the Information Technology Act, 2008. Also, the Act needed to be technology neutral to provide for alternative technology of electronic signature for bringing harmonization with model Law on Electronic signatures adopted by United Nations Commission on International Trade Law (UNCITRAL). To overcome all these problems new additions have been done by I.T. Amendment act 2008.

8.1 Causes:

Indian women citizens are still not open to immediately reporting the cyber abuse or cybercrime. The biggest problem of cybercrime lies in the modus operandi and the motive of the cyber-criminal. Cyberspace is a transit space for many people, including offenders. While people do not live in cyberspace, they come and go like any other place. This nature provides the offenders the chance to escape after the commission of cybercrime. Many websites and blogs provide security tips for the safety of women and children in the net. But still, then cybercrime against women are on rising.

In reality, it is seen many chat friends enjoy teasing their women friends with obscene words which are the virtual beginning of cyber obscenity. They slowly take their female friends into confidence and start discussing their own problems like a true friend. Hence, on many occasions they are successful in turning the net friendship into a strong bond and gradually proceed to send obscene or derogatory remarks. If the recipient shies away, the sender of such messages would become more encouraged to continue. The problem would be solved only when the victimized woman then and there reports back or even warn the abuser about taking strong actions.

8.2 Women as soft targets

With increasing traffic in the virtual world, the chances of falling prey to cybercrime loom large all the while, more so in the case of women who are often seen as soft targets.

In professional parlance, cybercrime covers a vast gamut of illegal online activities, such as fraud, money laundering, spam, identity theft, cyberstalking, drug trafficking, sexual predators, etc.

Mr. Pawan Duggal, an advocate specializing in cyber law in the Supreme Court once said, “I have observed that cybercrime and women in India have a deep relationship. My experience has been that women here by and large, shy away from reporting their matters, fearing potential negative media publicity, which may irreparably impact their reputations.”

More troubling is the rate at which these threats take place. Cybercriminals currently produces 20,000 to 30,000 new and unique pieces of malicious code every day in order to make it difficult for signature scanners to keep up.

8.3 Casting Couch:

Meaning and Reasons: The casting couch is a euphemism for a sociological phenomenon that involves the trading of sexual favors by an aspirant, apprentice employee, or subordinate to a superior, in return for entry into an occupation, or for other career advancements with an organization. Careers that are highly desirable and traditionally difficult to break into, such as the movie, television, and music industries, have been the subject of casting couch stories in popular culture. Such trading of favors is an abuse of power, and can become a wider sex scandal if deemed newsworthy.

The “casting couch” concept in many aspects is connotative of the entertainment industry, but may apply to any industry for which entry is highly desirable and competitive.

8.4 Eve-teasing:

Meaning: Eve teasing is a euphemism used in India for public sexual harassment, street harassment or molestation of women by men, with eve being a reference to the biblical eve. Eve-teasing has been a notoriously difficult crime to prove, as perpetrators often devise ingenious ways to attack women, even though many feminist writers

term it as “little rapes”, and usually occur in public places, streets, and public transport.

8.5 Causes of Eve teasing:

Bad intentions such as revenge, ego, the high altitude of sexual desire, etc.

Effect of cinema

Psychological problems

Legal redress:

Victims usually seek recourse through 298 (A) and (B) of the Indian Penal Code (IPC), which sentences a man found guilty of making a girl or woman the target of obscene gestures, remarks, songs or recitation for a maximum term of three months.

Section 292 of the IPC clearly spells out that showing pornographic or obscene pictures, books or slips out that showing pornographic or obscene pictures, books or slips to a woman or girl draws a fine of Rs. 2000/- with two years of rigorous imprisonment for first offenders. In case of repeated offence, the offender may have a fine Rs.5000/- with five years imprisonment imposed.

9. Conclusion:

The growth of cybercrime in India, as all over the world, is on the rise. Anybody who uses the internet is at risk of becoming a victim of cybercrime. Cyber space offers a plethora of opportunities for cybercriminals either to cause harm to innocent people.

India is considered as one of the very few countries to enact the IT Act 2000 to combat cybercrimes. This Act is widely covered commercial and economic crimes which is clear from the preamble of the IT Act but it is observed that there is no specific provision to protect the security of women and children. However, there are few provisions to cover some of the crimes against women in cyberspace under IT Act.

In order to avoid the cybercrime, we should not engage in conversation with people we don't know. People on the other end of the computer may not be who they claim to be. We must keep our passwords protected and do not keep sensitive material on the computer as that can be accessed by the hacker. If anything seems out of place or wrong, contact law enforcement immediately.

Indian women netizens are still not open to immediately report the cyber abuse or cyber-crime. This nature provides the offenders the chance to escape after the commission of cyber-crime. The problem would be solved only when the victimized women then and there report back or even warn the abuser about taking strong actions.

References:

1. Indian Penal code, 1860
2. Laws and legislative measures affecting women by National commission for Women (NCW)
3. <http://en.wikipedia.org>
4. Indian feminism by Geetanjali Gangoli.
5. The times of India
6. The Hindu
7. NCRB (National crimes Record Bureau)
8. www.Tribune ndia.com
9. Cyber Crime cell, Andhra Pradesh
10. Cyberlawsindia.net
11. UNDP, Voices and Rights, March,2010
12. Jagori—Revisiting Dowry in India,2009
13. Madhu Kishwar, Challenges of Democracy, QUP,2006
14. Thaindian.com/news portal
15. Annamtani - 'violence and women's city '-2000
16. Indian express-CNN-IBN-CSDS
17. Women Speak (2000)-United voices against poverty and violence in India.
18. Keeping women safe ...By Rohini Pandey.
19. Rethinking the Challenge of Women's safety in India's cities-Meher Soni.
20. Kavitha Krishnan- "Don't need big brother watching us..."



PHARMACEUTICAL CRIMES: TRENDS, PATTERNS AND CHALLENGES IN INDIA

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Abstract

The abuse of Pharmaceutical Drugs in India has assumed serious problems. The problem seems to be extended in the regions of North-East, North-West and South-West of the country. These problems are about synthetic drugs that are Tablets, Capsules, Syrups and Injections. The pharmaceutical crimes come under the Drugs and Cosmetics Act 1940 and Rules 1945 and the Narcotic Drugs and Psychotropic Substances (NDPS) Act, 1985. The objective of the study is to explore pharmaceutical crimes in India. Through this paper, an attempt has been made to address the trends, patterns and challenges of pharmaceutical crimes in our country. The present study is based on the exploratory research design. The data sources are from the Narcotics Control Bureau, Annual Report published by the Ministry of Home Affairs, Government of India. The data has been analyzed statistically from the year 2016 to 2018. Despite strict controls and monitoring the sale, purchase, storage, etc., of pharmaceutical crimes, there is evidence indicating diversion of pharmaceutical substances. The researcher has addressed the recent trends, patterns and challenges indicating that synthetic drugs are now replacing natural and semi-synthetic drugs based on the findings. Hence, coordination (national & international) and concerted efforts are the need of the hour to counter the misuse and abuse of pharmaceutical crimes effectively in India.

Keywords: Pharmaceutical Crimes, Trends, Patterns, Challenges and India.

Introduction

Pharmaceutical preparations of Narcotic Drugs and Psychotropic Substances come under the purview of Drugs and Cosmetics Act 1940 and Rules 1945 and Narcotic Drugs and Psychotropic Substances (NDPS) Act, 1985. The Narcotic Drugs and Psychotropic Substances (NDPS) Act, 1985 was framed taking into account India's obligations under the three United Nations Conventions. The single convention on Narcotic Drugs, 1961 as amended by the 1972 protocol, the Convention on Psychotropic Substances, 1971, the Convention against illicit Traffic in Narcotic Drugs and Psychotropic Substances, 1988 and the South Asian Association for Regional Cooperation (SAARC) Convention, 1993 (AR-NCB, 2017) as well as Article 47 of the Constitution of

India which guidelines that the “state shall endeavour to bring about prohibition of the consumption, except for medical purposes, of intoxicating drinks and of drugs which are injurious to health”.

The act prohibits, except for medical or scientific purposes, the manufacture, production, trade, use and so forth of Narcotic Drugs and Psychotropic Substances. Unlike the earlier Opium Acts, 1857 and 1878 and the Dangerous Drugs Act, 1930 which are replaced by the Narcotic Drugs and Psychotropic Substances (NDPS) Act, 1985 which came into effect from 14th November 1985 expressly provided constitution of a central authority to exercise the powers and functions of the Central Government under the act. In the exercise of these powers, the Narcotics Control Bureau (NCB) was constituted with Headquarters at New Delhi with effect from 17th March 1986 (AR-NCB, 2016).

The Government of India, Ministry of Home Affairs, Internal Security Division-II (Narcotics Control Bureau) Ministry of Finance, Department of Revenue (Central Bureau of Narcotics) Ministry of Health and Family Welfare, Directorate General of Health Services (Central Drugs Standard Control Organization) & Departments of Health and Family Welfare (Drug De-addiction Programme) Ministry of Social Justice and Empowerment, Department of Social Justice and Empowerment (Social Defence) and Ministry of Chemical and Fertilizers, Department of Chemicals and Petrochemicals Industry (Chemicals Division) are making earnest endeavors towards controlling the drugs and handling the difficulties of unlawful cultivation and collection of Opium, Cannabis and Coca leaf, processing of Narcotic Drugs, Psychotropic Substances, Precursor Chemicals and Pharmaceutical Products of various forms like Tablets, Capsules, Syrups and Injections.

Objective

The objective of the study is to explore pharmaceutical crimes in India. Through this paper, an attempt has been made to address the trends, patterns and challenges of pharmaceutical crimes in our country.

Methodology

The present study is based on an exploratory research design. The data is taken from NCB and MHA reports. The data has been analyzed statistically from the year 2016 to 2018.

Results and Analysis

Pharmaceutical crimes

The abuse of pharmaceutical drugs in India has assumed serious problems. The problem seems to be extended in the regions of North-East,

North-West and South-West of the country. These problems are about synthetic drugs that are Tablets, Capsules, Powders, Syrups and Injections. The pharmaceutical products are Amphetamine Type Stimulants (ATS), New Psychoactive Substances (NPS), Benzodiazepines, Codeine-based cough syrups and Ketamine injections. The easy availability of such preparations is the major factor that encourages their misuse. There is also a perception that these pharmaceutical drugs are less harmful than semi-synthetic and natural drugs like Opium, Heroin, Cocaine etc.

- Amphetamine Type Stimulants (ATS) are Amphetamine, Methamphetamine, Methcathinone, Methaqualone, Mephedrone, Ecstasy-group substances (e.g., 3,4-Methylenedioxymethamphetamine (MDMA) and its analogues).
- New Psychoactive Substances (NPS) are Acetylfentanyl, N-Benzylpiperazine, Methyone, MDPV (3,4-Methylenedioxypropylvalerone), Para-methoxymethylamphetamine (PMMA), Pyrrolidinovalerophenone (PVP), Para methyl -4-methylaminorex 4,4' DMAR, Methoxetamine, Phenazepam and Ketamine.
- Benzodiazepines are Alprazolam, Diazepam, Lorazepam, Clonazepam and Zolpidem.
- Codeine-based cough syrups are Corex, Phenstasyl and Recodex.

Pharmaceutical crimes spread the accompanying criminal offences, independent of which National Legislation the wrongdoing falls under: The manufacturing and distribution of fake or misrepresented (misleading/counterfeit/erroneously marked) pharmaceuticals through lawful to unlawful supply chains including: robbery, misrepresentation, redirection, sneaking, illicit exchange, illegal tax avoidance, narcoterrorism, web drug store, digital money and bitcoin. Driven by financial profit, criminal components have turned out to be engaged with all aspects of the chain of supply of illegal medicines, from manufacturing to distribution. To turn significant benefits, criminals are typically required to have a high level of refinement and association. However, pharmaceutical drugs and crimes are generally not understood to be as organized as more established criminal activities, such as drug trafficking or smuggling (Interpol, 2014).

Tablets

Tablets are solid preparation each containing a single dose of one or more active substances and are usually obtained by compressing a uniform volume of particles. Tablets triturates are prepared by using a hand moulding machine and a compressed tablet machine.

Table: 01. Details of Tablets seized in India from 2016 to 2018.

Tablets (No's)	Details	2016	2017	2018
	Quantities	7961039	6572350	88595264 + 47.32 kg
	Cases	2046	3219	•
	Arrests	2238	3611	•

(• = Not Reported)

Source: Data released by Director General, Narcotics Control Bureau, Ministry of Home Affairs, Government of India. Annual Report - 2016, p. 85-86, 2017, p. 84-85 and 2018, p. 80.

The number of Tablet seizures in India has luctuated over the period 2016 to 2018, and the highest reported seizures in 2018 were 88595264 + 47.32 kg. Tablet seizures in 2018 were 88595264 + 47.32 kg. The comparative figures for the last three years are presented at Table 01. While there has been an increasing trend in the number of reported Tablet seizures compared to the previous year 2017, the number of tablets seized has increased from 6572355 to 88595264 + 47.32 kg.

The number of Tablet cases in India has fluctuated over the period 2016 to 2018 and the highest reported cases in 2017 were 3,219. The comparative figures for the last three years are presented at Table 01. While there has been an increasing trend in the number of reported cases compared to the previous year 2016, the number of Tablet cases has increased from 2046 to 3219. In the year 2018, the number of Tablet cases was not reported.

The number of Tablet arrests in India has fluctuated over the period 2016 to 2018 and the highest reported arrests in 2017 were 3611. Tablet arrests in 2017 were 3611. While there has been an increasing trend in the number of reported arrests compared to the previous year 2016, the number of Tablet arrests has increased from 2238 to 3611. In the year 2018, the number of Tablet arrests was not reported.

Table: 02. Details of Tablet Seizures, Cases and Arrests in Region wise in India from 2016 to 2018.

Region	Details	2016	2017	2018
North Region (NR)				
	Quantity	2136878	5058593	81260983 + 8.87 kg
	Cases	1814	2874	•
	Arrests	1904	3134	•

East Region (ER)	Quantity	2624938	1421689	5745766 + 27.85 kg
	Cases	181	273	•
	Arrests	262	377	•
South West Region (SWR)	Quantity	3199223	92068	1588515 + 10.6 kg
	Cases	51	72	•
	Arrests	72	100	•
Totals	Quantity	7961039	6572350	88595264 + 47.32 kg
	Cases	2046	3219	•
	Arrests	2238	3611	•

(• = Not Reported)

The number of Tablet seizures, cases and arrests in India has increased over the period 2016 to 2018. The North Region seized the Tablet in 2018 the amount of Tablet was 81260983 + 8.87 kg, and the seizures happened in Chandigarh, Haryana, Himachal Pradesh, Jammu & Kashmir, New Delhi, Punjab, Rajasthan, Uttar Pradesh and Uttarakhand. In East Region seized more quantity of Tablet in 2018 the amount of Tablet was 5745766 + 27.85 kg, and the seizures happened in Arunachal Pradesh, Assam, Bihar, Chhattisgarh, Jharkhand, Manipur, Meghalaya, Mizoram, Nagaland, Odisha, Sikkim, Tripura and West Bengal. In South-West Region seized the Tablet in 2016 the number of Tablet was 3199223 and the seizures happened in Andhra Pradesh, Gujarat, Kerala, Madhya Pradesh, Maharashtra, Tamil Nadu and Telangana. Tablet seizures in 2018 were 88595264 + 47.32 kg. The comparative tables for the last three years are presented at Table 02. While there has been an increasing trend in the number of reported Tablet seizures compared to the previous year 2017, the quantity of Tablet seized has increased from 6572350 to 88595264 + 47.32 kg. The number of Tablets seized more quantity in the North Region (88456454 + 8.87 kg) than compared to the East Region (9792393 + 27.85 kg), and the South-West Region (4879806 + 10.6 kg). The seized quantity of Tablets in the North Region increased 5058593 to 81260983 + 8.87 kg, the East Region increased 1421689 to 5745766 + 27.85 kg, and the South-West Region increased 92068 to 1588515 + 10.6 kg it depending on the drug trafficking.

The highest cases were in North Region in 2017 the number of cases was 2874 followed by the East Region in 2017 the number of cases was 273 and in the South-West Region in 2017 the number of cases was 72. Tablet cases in

2017 were 3219. The comparative tables for the last three years are presented at Table 02. While there has been an increasing trend in the number of reported Tablet cases compared to the previous year 2016, the number of cases has increased from 2046 to 3219. In the year 2018, the number of Tablet cases was not reported.

The highest arrests were reported in North Region in 2017 the number of arrests was 3134 followed by the East Region in 2017 the number of arrests was 377 and the South-West Region in 2017 the number of arrests was 100. Tablet arrests in 2017 were 3611. The comparative tables for the last three years are presented at Table 02. While there has been an increasing trend in the number of reported Tablet arrests compared to the previous year 2016, the number of arrests has increased from 2238 to 3611. In the year 2018, the number of Tablet arrests was not reported.

Syrups

Syrup is a concentrated or nearly saturated solution of sucrose in purified water. The concentration of sugar is 66.7% w/w. The syrups are sweet viscous preparations. The syrups containing medicinal substances are called “medicinal syrup” and those containing aromatic or flavored substances are known as “flavored syrups”.

Table: 03. Details of Syrups seized in India from 2016 to 2018.

Syrups (No's)	Details	2016	2017	2018
	Quantities	902600	566450	877838
	Cases	428	568	•
	Arrests	553	609	•

(● = Not Reported)

Source: Data released by Director General, Narcotics Control Bureau, Ministry of Home Affairs, Government of India. Annual Report - 2016, p. 85-86, 2017, p. 84-85 and 2018, p. 80.

The number of Syrup seizures in India has fluctuated over the period 2016 to 2018 and the highest reported seizures in 2016 were 902600. Syrup seizures in 2018 were 877838. The comparative figures for the last three years are presented at Table 03. While there has been an increasing trend in the number of reported Syrup seizures compared to the previous year 2017, the number of Syrup seizures has increased from 566450 to 877838.

The number of Syrup cases in India has fluctuated over the period 2016

to 2018 and the highest reported cases in 2017 were 568. Syrup cases in 2017 were 568. The comparative figures for the last three years are presented at Table 03. While there has been an increasing trend in the number of reported cases compared to the previous year 2016, the number of Syrup cases has increased from 428 to 568. In the year 2018, the number of Syrup cases was not reported.

The number of Syrup arrests in India has fluctuated over the period 2016 to 2018 and the highest reported arrests in 2017 were 609. Syrup arrests in 2017 were 609. The comparative figures for the last three years are presented at Table 03. While there has been an increasing trend in the number of reported arrests compared to the previous year 2016, the number of Syrup arrests has increased from 553 to 609. In the year 2018, the number of Syrup arrests was not reported.

Table: 04. Details of Syrup Seizures, Cases and Arrests in Region wise in India from 2016 to 2018.

Region	Details	2016	2017	2018
North Region (NR)	Quantity	41335	170037	47348
	Cases	246	397	
	Arrests	328	399	
East Region (ER)	Quantity	840809	309560	233106
	Cases	170	143	
	Arrests	201	166	
South West Region (SWR)	Quantity	20456	86853	597384
	Cases	12	28	
	Arrests	24	44	
Totals	Quantity	902600	566450	877838
	Cases	428	568	
	Arrests	553	609	

(●= Not Reported)

The number of Syrup seizures, cases and arrests in India has increased over the period 2016 to 2018. The East Region seized more quantities of Syrup in 2016 the amount of Syrup was 840809, and the seizures happened in Assam, Bihar, Chhattisgarh, Jharkhand, Manipur, Meghalaya, Mizoram, Nagaland, Odisha and West Bengal. In North Region seized the Syrup in 2017 the amount of Syrup was 170037, and the seizures happened in Chandigarh, Himachal Pradesh, Jammu & Kashmir, New Delhi, Punjab and Rajasthan. In South-West

Region seized the Syrup in 2018 the amount of Syrup was 597384, and the seizures happened in Gujarat, Karnataka, Madhya Pradesh and Maharashtra. Syrup seizures in 2018 were 877838. The comparative tables for the last three years are presented at Table 04. While there has been an increasing trend in the number of reported Syrup seizures compared to the previous year 2017, the quantity of Syrup seized has increased from 566450 to 877838. The number of Syrup seizures more quantity in the East Region (1383475) than compared to the South-West Region (704693) and the North Region (258720). The seized quantity of Syrup in the South-West Region increased 86853 to 597384, the East Region reduced 309560 to 233106, the North Region reduced 170037 to 47348 it depends on the drug trafficking.

The highest cases were in North Region in 2017 the number of cases was 397 followed by the East Region in 2016 the number of cases was 170 and in the South-West Region in 2017 the number of cases was 28. Syrup cases in 2017 were 568. The comparative tables for the last three years are presented at Table 04. While there has been an increasing trend in the number of reported Syrup cases compared to the previous year 2016, the number of cases has increased from 428 to 568. In the year 2018, the number of Syrup cases was not reported.

The highest arrests were reported in North Region in 2017 the number of arrests was 399 followed by the East Region in 2016 the number of arrests was 201 and in the South-West Region in 2017 the number of arrests was 44. Syrup arrests in 2017 were 609. The comparative tables for the last three years are presented at Table 04. While there has been an increasing trend in the number of reported Syrup arrests compared to the previous year 2016, the number of arrests has increased from 553 to 609. In the year 2018, the number of Syrup arrests was not reported.

Injections

Parenteral products are otherwise known as Injections. Injections are sterile preparations administered parenterally. Injections are administered either Intravenously, Intramuscularly, Subcutaneously, Intradermally, Intraspinally or Intrathecally.

Table: 05. Details of Injections seized in India from 2016 to 2018.

Injection (No's)	Details	2016	2017	2018
	Quantities	418513	58463	•
	Cases	176	485	•
	Arrests	206	763	•

(● = Not Reported)

Source: Data released by Director General, Narcotics Control Bureau, Ministry of Home Affairs, Government of India. Annual Report - 2016, p. 85-86, 2017, p. 84-85 and 2018, p. 80.

The number of Injection seizures in India has fluctuated over the period 2016 to 2018, and the highest reported seizures in 2016 were 418513. Injection seizures in 2017 were 58463. The comparative figures for the last three years are presented at Table 05. While there has been a decreasing trend in the number of reported Injection seizures compared to the previous year 2016, the number of Injection seizures has decreased from 418513 to 58463. In the year 2018, the number of Injection seizures was not reported.

The number of Injection cases in India has fluctuated over the period 2016 to 2018, and the highest reported cases in 2017 were 485. Injection cases in 2017 were 485. The comparative figures for the last three years are presented at Table 05. While there has been an increasing trend in the number of reported cases compared to the previous year 2016, the number of Injection cases has increased from 176 to 485. In the year 2018, the number of Injection cases was not reported.

The number of Injection arrests in India has fluctuated over the period 2016 to 2018 and the highest reported arrests in 2017 were 763. Injection arrests in 2017 were 763. The comparative figures for the last three years are presented at Table 05. While there has been an increasing trend in the number of reported arrests compared to the previous year 2016, the number of Injection arrests has increased from 206 to 763. In the year 2018, the number of Injection arrests was not reported.

Table: 06. Details of Injection Seizures, Cases and Arrests in Region wise in India from 2016 to 2018.

Region	Details	2016	2017	2018
North Region (NR)	Quantity	36550	48348	•
	Cases	152	456	•
	Arrests	165	730	•
East Region (ER)	Quantity	26329	4864	•
	Cases	13	14	•
	Arrests	13	14	•
South West Region (SWR)	Quantity	355634	5251	•

	Cases	11	15	•
	Arrests	28	19	•
Totals	Quantity	418513	58463	•
	Cases	176	485	
	Arrests	206	763	

(● = Not Reported)

The number of Injection seizures, cases and arrests in India has increased over the period 2016 to 2018. The South-West Region seized more quantity of Injections in 2016 the number of Injection was 355634, and the seizures happened in Kerala, Madhya Pradesh and Tamil Nadu. In North Region seized the Injection in 2017 the number of Injection was 48348 and the seizures happened in Chandigarh, Himachal Pradesh, Jammu & Kashmir, New Delhi, Punjab, Uttar Pradesh and Uttarakhand. In East Region seized the Injection in 2016 the number of Injection was 26329, and the seizures happened in Assam, Chhattisgarh, Mizoram and West Bengal. Injection seizures in 2017 were 58463. The comparative tables for the last three years are presented at Table 06. While there has been a decreasing trend in the number of reported Injection seizures compared to the previous year 2016, the number of Injection seizures has decreased from 418513 to 58463. The number of Injection seizures is more quantity in the South-West Region (360885) than compared to the North Region (84898) and the East Region (31193). The seized quantity of Injection in the North Region increased 36550 to 48348, the East Region reduced 26329 to 4864, and the South-West Region reduced 355634 to 5251 it depends on the drug trafficking. In the year 2018, the number of Injection seizures was not reported.

The highest cases were in North Region in 2017 the number of cases was 456 followed by the South-West Region in 2017 the number of cases was 15 and in the East Region in 2017 the number of cases was 14. Injection cases in 2017 were 45. The comparative tables for the last three years are presented at Table 06. While there has been an increasing trend in the number of reported Injection cases compared to the previous year 2016, the number of cases has increased from 176 to 485. In the year 2018, the number of Injection cases was not reported.

The highest arrests were reported in North Region in 2017 the number of arrests was 730 followed by the South-West Region in 2016 the number of arrests was 28 and in the East Region in 2017 the number of arrests was 14. Injection arrests in 2017 were 730. The comparative tables for the last three years are presented at Table 06. While there has been an increasing trend in the number of reported Injection arrests compared to the previous year 2016, the number of arrests has increased from 206 to 763. In the year 2018, the number of Injection arrests was not reported.

Discussion

The drug trafficking scenario in India is largely attributed to various internal and external factors. One of the internal factors influencing drug trafficking in India is the manufacture of a lot of synthetic drugs and precursor chemicals that are smuggled out of the country. The external factor is the prime geographical factor influencing India's drug situation as it is near the regions of South West and South East Asia known as a "Golden Crescent" (Afghanistan, Pakistan and Iran) and the "Golden Triangle" (Myanmar, Laos and Thailand). India's geographical location makes it vulnerable to transit, trafficking and consumption of Narcotic Drugs, Psychotropic Substances, Precursor Chemicals and Pharmaceutical Products of various forms like Tablets, Capsules, Syrups, and Injections.

The major trafficking of Tablets mainly takes place from North Region (Chandigarh, Haryana, Himachal Pradesh, Jammu & Kashmir, New Delhi, Punjab, Rajasthan, Uttar Pradesh and Uttarakhand), South-West Region (Andhra Pradesh, Gujarat, Kerala, Madhya Pradesh, Maharashtra, Tamil Nadu and Telangana), and East Region (Arunachal Pradesh, Assam, Chhattisgarh, Manipur, Meghalaya, Mizoram, Nagaland and West Bengal). Traditionally, precursor chemicals have been trafficked from India to Myanmar whereas the finished Amphetamine Type Stimulants product has been trafficked in the reverse direction. The seizures of Amphetamine and Methamphetamine pills in India are predominantly made in the North-Eastern Border States and one of the important trafficked through courier parcels. Whereas Amphetamine and Methamphetamine pills are mostly trafficked into the country from abroad, Amphetamine and Methamphetamine in powder form are primarily manufactured in India. The seizures of Methamphetamine continue to follow upward trends, both in terms of Tablet and in crystalline form (GSP-UNODC, 2012).

The Narcotic Drugs and Psychotropic Substances Act, 1985 (61 of 1985) amendment of Tramadol as a Psychotropic Substances S. O. 1761(E) with effect from 26th April 2018. The impact of the amendment, the Indian Drug Manufacturers Association wrote the letter to the Department of Revenue, Ministry of Finance requested that the minimum required six months be given for ensuring proper compliance with the amended order/rules. Rules 65 may be amended to include a proviso, stating "Any new substance which has an established medical or scientific use and included in the list of psychotropic substances specified in the schedule to the Act would come into effect on the 180th day from the date of such notification". Based on the request, the Narcotic Drugs and Psychotropic Substances Act, 1985 (61 of 1985) amendment S. O. 3448(E) with effect from 13th July 2018. The licensed manufacture, importers and exporters of Tramadol shall be covered under the provisions of this notification after the expiry of a period of 120 days from the date of its publication in the Official Gazette.

The major trafficking of Syrups mainly takes place from East Region (Assam, Bihar, Chhattisgarh, Jharkhand, Manipur, Meghalaya, Mizoram, Nagaland, Odisha and West Bengal), North Region (Chandigarh, Himachal Pradesh, Jammu & Kashmir, New Delhi, Punjab and Rajasthan), and South-West Region (Madhya Pradesh and Maharashtra). Trafficking of Codeine Based Cough Syrups is mainly along the border, especially at Indo-Bangladesh Border. Codeine based cough syrups are mainly trafficked from Northern states like Uttar Pradesh, New Delhi, Bihar to North-Eastern states by road in trucks and railway parcels and there to Bangladesh. Codeine based cough syrups brands like Corex, Phensedyl and Recodex are abused and trafficked. In recent times, illicit manufacture, diversion, storage, and trafficking of Codeine based cough syrups, which got developed in the process of continuous efforts of Narcotics Control Bureau, North Region, Zonal Unit of Lucknow seized 60971 bottles of Phensedyl Syrup bearing the brand name Phensedyl made by Abbott company. While the cough syrups were loaded in a truck in the late evening on 17th January 2021 from a rented Godown. In this case, six persons were arrested and seized two trucks.

The major trafficking of Injections mainly takes place from South-West Region (Kerala, Madhya Pradesh and Tamil Nadu), North Region (Chandigarh, Himachal Pradesh, Jammu & Kashmir, New Delhi, Punjab, Uttar Pradesh and Uttarakhand), and East Region (Assam, Chhattisgarh, Mizoram and West Bengal). Ketamine is the list of New Psychotropic Substances, under the Narcotic Drugs and Psychotropic Substances Act, 1985 vide its notification dated on 10th February 2011. In recent years, manufacture and trafficking cases indicate that Indian drug trafficking syndicates may be playing a role in the Ketamine market in the region. In September 2016, officers of Mizoram Excise and Narcotics seized 19843 vials of Ketamine at Zarkawt, Aizawl. Two persons were arrested. The suspected destination of the seized drug was Myanmar (AR-NCB, 2016). Mephedrone is the list of New Psychotropic Substances, under the Narcotic Drugs and Psychotropic Substances Act, 1985 vide its notification dated 05th February 2015. The first seizure of an illicit Mephedrone was reported in February 2016 in Pune. In this case, 0.170 kg of Mephedrone was sized and two persons were arrested (AR-NCB, 2016).

The chemistry of pharmaceutical crime occurs when there is an intersection in time and space. The area of research is an ongoing debate in the contemporary world; it has overlooked three extremely important points illicit manufacture drugs, illicit drug trade and illicit financial flows. Pharmaceutical crime is likely to occur when three elements interlink between illicit manufacture, illicit trade and illicit financial flows (see Figure 01). The first element is the illicit manufacturing process of clandestine laboratories or factories to the production of pharmaceutical drugs or medications. Second, the illicit trade process of the distribution wholesales and retails. Third, illicit financial flows of digital currency like bitcoin.

Figure. 01. A Concept of the Pharmaceutical Crime



Recommendations

- To control the Precursor Chemicals (Import & Export).
- To control the Pharmaceutical Drugs (Tablets, Syrups & Injections).
- To control the Clandestine laboratories & factories.
- To need pharmacy experts to deal with Pharmaceutical Crimes.
- To need cyber-experts to deal with Internet Pharmacy Crimes.
- To need forensic experts to the analysis of drugs.
- To need separate research wing for Pharmaceutical crimes.
- To increase the capacity of officers by NCB, CBN & Customs.
- To need Training programs for other law enforcement agencies and stakeholders.
- To strengthen the laws in regulating Pharmaceutical products diversion.
- To call research proposal related to NDPS research & provide funds for research.
- To need more joint operation (National & International).
- To maintain Border Management North-East and North-West and South-West regions of India.

Conclusion

With the strict controls and monitoring of the sale, purchase, storage, etc., of certain drugs, there was evidence indicating diversion of pharmaceutical drugs.

Drug law enforcement officers should keep track of the current drug trafficking trends, patterns, and challenges of legitimate to illegitimate pharmaceutical drugs. Based on the analysis, the researcher has addressed the recent trend indicating that seizures, cases and arrests are increased. The ongoing trend phenomenal development of countless synthetic drugs of redirection precursor chemicals has fundamentally tested our comprehension of synthetic drugs and increased the use of the internet, the prevalence of illicit Erectile Dysfunction (ED) medication, the prevalence of doping substances, illicit medicines and narcotics, tenuous ties to terrorism. The new pattern indicates the unlawful manufacture of synthetic drugs, for example, the issue of undercover research facilities fabricating Amphetamine Type Stimulants (ATS), New Psychoactive Substances (NPS), and Benzodiazepines. A new set of challenges has emerged for monitoring, drug detection and identification, legislative measures, and drug control. The complexity and diversity of pharmaceutical drugs pose a significant challenge for law enforcement and international control efforts. The national coordination with other agencies like IB, RAW, DRI, CBN, Indian Coast Guard, ED, C&CE, FIU, BSF, SSB, ITBP, CISF, CRPF, RPF, AR, various State Police and international coordination with SAARC and UN councils and concerted efforts are the need of the hour to counter the misuse and abuse of pharmaceutical drugs effectively in India.

References

1. Annual Report. 2016. Narcotics Control Bureau, Ministry of Home Affairs. New Delhi: Government of India.
2. Annual Report. 2017. Narcotics Control Bureau, Ministry of Home Affairs. New Delhi: Government of India.
3. Annual Report. 2018. Narcotics Control Bureau, Ministry of Home Affairs. New Delhi: Government of India.
4. Article 47 of the Constitution of India.
5. Global SMART Programme. (2012). Patterns and Trends of Amphetamine Type Stimulants and Other Drugs: Asia and the Pacific. Austria: United Nations Office on Drugs and Crime.
6. Interpol. 2014. Pharmaceutical crime and Organized criminal groups; An analysis of the involvement of organized criminal groups in pharmaceutical crime since 2008. France. Retrieved from www.interpol.int.
7. Mithal B M. (2nd Ed.) 2007. A Textbook of forensic pharmacy. New Delhi: Vallabh Prakashan.



Every Contact Leaves a Trace: The DNA

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Abstract

From laboratory to courtroom, DNA is one such piece of evidence that is a witness in itself. Blood, semen and saliva along with other circumstantial evidences play a major role in the conviction of the perpetrator in sexual assault (SA) cases. Biological evidence (BE) helps in linking the perpetrator with the victim and the crime scene. These are of particular use in Sexual Offence (SO) cases as they are the source for the genetic material that can place a person in the category of accused or innocent based upon the matching profile. Many times, BE remains the only way to prove the sexual contact between the victim and the perpetrator and the correct identification of the perpetrator. This paper focuses on the relevance of BE in cases of SO and the guidelines to help forensic, health and law experts to deal with the BE for DNA analysis. Such evidence must be carefully managed to avoid any contamination, degradation, or loss as the relevance of any finding in Forensic Genetics is based on, in the first instance, the integrity and quantity of the samples submitted for analysis.

Keywords: Crime scene, Biological evidence, DNA, Investigation, Sexual offence.

Introduction

“It is impossible for a criminal to act, especially considering the intensity of a crime, without leaving traces of his presence.” This line once said by Edmond Locard lays the foundation of Forensic Science today. “Every contact leaves a trace’ is the fundamental Forensic principle. The same rule applies in the case of SOs also.

Our law criminalizes a broad array of sexual and sex-related conduct. It includes rape, SA, coercion, human trafficking, female genital mutilation, forced marriage, sexual humiliation, voyeurism, public nudity and public indecency, sexual transmission of disease, selling and buying sexual services (prostitution), pimping and pandering, statutory rape and child molestation, abuse of a position

of trust, child grooming, creating and possessing child pornography, revenge porn, failure to register as a sex offender, fornication, sodomy, adultery, assault by sadomasochism, adult and child incest, bigamy, polygamy, bestiality, necrophilia and sale of sex toys [1].

Though there are a number of SOs and all are heinous crimes, the most highlighted among all is “RAPE”. This is so because the people are not even aware of some of the acts that come under the category of SOs, in others, there is no reporting of cases due to many societal dogmas, fear, insecurities, family pressure, etc.

According to data published by UN WOMEN [2],

- Globally, 35 percent of women have ever experienced physical and/or sexual intimate partner violence or sexual violence by a non-partner.
- At least 155 countries have passed laws on domestic violence and 140 have laws on sexual harassment in the workplace.
- 15 million adolescent girls worldwide, aged 15–19 years, have experienced forced sex”.

There is a wide gap between the commission and the reporting of SOs. Out of all, the most reported cases include rape cases.

According to a report by Thomson Reuters Foundation (TRF), India is perceived to be the most unsafe country for women due to the rising cases of SAs in the country. But the data from National Family Health Survey (NFHS) conducted across developing nations of the world shows that the incidence of sexual violence experienced by women is lower in India as compared to most other countries for which data are available and the incidence of such violence has declined over time (Figure 1) [3].

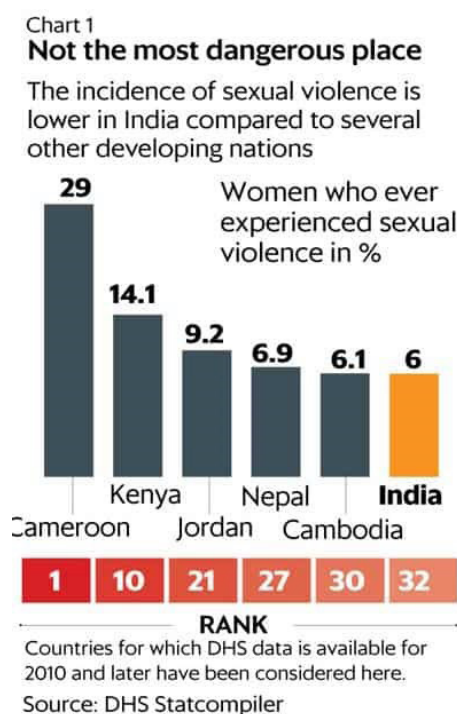


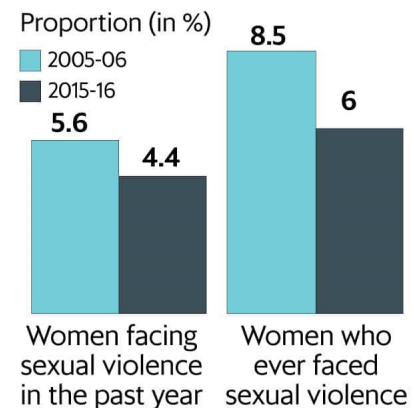
Figure 1: Rank of some of the countries and incidences of SOs in terms of percentage of women who ever experienced sexual violence

The Demographic Health Survey (DHS), also known as the NFHS in India, interviewed more than 700,000 women in 2015-16 during the last round of the survey.

Most of the SA cases go unreported in country due to high number of cases involving any offender known to the victim. Even among crimes that are reported, most perpetrators are known to the victims. But the actual incidence of SAs has declined over the past decade, as recorded by NFHS, even as the reporting of SAs has risen. The rise in reported cases of sexual violence has been particularly sharp after the infamous Delhi gang rape of 2012, which led to increased awareness on this issue (**Figure 2** and **3**) [3].

Chart 3A

Sexual violence in the country has declined over the past decade...

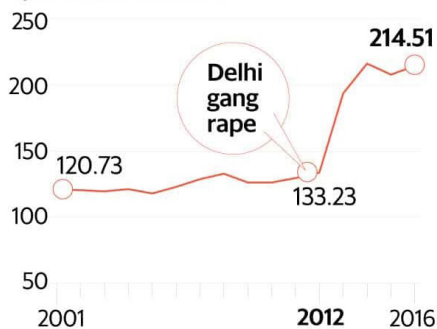


Source: DHS Statcompiler

Figure 2: Decline in sexual violence cases in the past decade

...Even as the reporting of sexual violence has increased, especially after 2012

Reported rate of sexual violence (per million females)



Source: NCRB, Census

Figure 3: Rise in reporting of SOs after 2012

It can be concluded from the above data that the report of TRF is likely due to the increased reporting of sexual violence in the country over the past few years [3].

Forensic Significance of Biological Evidences in Cases of SO

DNA is the most powerful investigative tool when it comes to SOs. With the exception of identical twins, no two people have the same DNA. This makes DNA, a highly reliable piece of evidence in SA cases. DNA evidence collected from a crime scene can be linked to a suspect or can eliminate a suspect from suspicion. During a SA, for example, biological evidence such as hair, skin cells, semen or blood may be left on the victim's body or at the crime scene. Properly collected DNA can be compared with known samples to place a suspect at the scene of the crime. Figure 4 showcases the collection of different possible evidence in SA cases and their probative value in individualizing a perpetrator.

Source: *Biological Evidence Analysis in Cases of SA* [4]

The semen, saliva, and blood are the most common and important BE that are encountered in any SO.

SEMEN – Semen works as a potent forensic marker to link a male perpetrator with the female victim and the crime scene in cases involving sexual acts. Semen is highly fluorescent and also contains spermatozoa. Semen can be found from clothes, victim's body, perpetrator, crime scene, etc. There might be a case where the perpetrator may be vasectomized or azoospermic. But the latest technologies

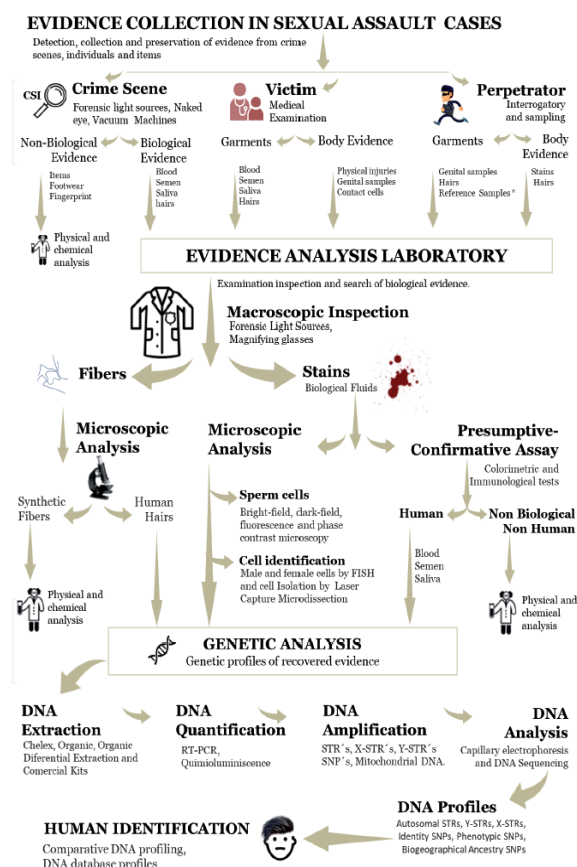


Figure 4: Evidence collection and analysis in SA cases

like FISH and Y-STR [5-7] can still help in linking the male perpetrator with the crime. Prostate-Specific Antigen (PAS) test is used for antigen grouping in the case of azoospermic or vasectomized males [8].

BLOOD – Serum antigen and ABO group analysis can help in the identification of perpetrator. DNA can also be isolated from blood and so the bloodstains on the body and clothes of the perpetrator or the victim play a major role in linking the victim, the crime scene, and the perpetrator [8].

SALIVA – There is a high chance of the perpetrator kissing, sucking or biting the victim which leads to transfer/deposition of oral fluid on the body of the victim. To trace saliva, not only the body of the victim but also nearby cigarette butts, cans, bottles, glass and even the clothes of the victim can be analyzed. Also, there is the presence of epithelial cells in saliva which helps in DNA analysis, therefore, helping in identifying the identity of the perpetrator [8].

For a good recovery and analysis of any of the biological evidence, the victim must report the assault as soon as possible and get the forensic medical examination (FME) done. The investigating officer has to make sure that the victim does not wash his/her face or any body parts where chances of finding BE are high.

A look at the “witness in itself: the DNA”

It was in 1986 when DNA was first used in a criminal investigation by Dr. Alec Jeffreys. In India, it was Dr. Lalji Singh also known as the Father of DNA Fingerprinting in India whose efforts took DNA fingerprinting to the mainstream in India, both in terms of research and forensic applications. Encouraged by his work, the Government of India gave him the task of establishing a Centre for DNA Fingerprinting and Diagnostics (CDFD) at Hyderabad in the late 1990s.

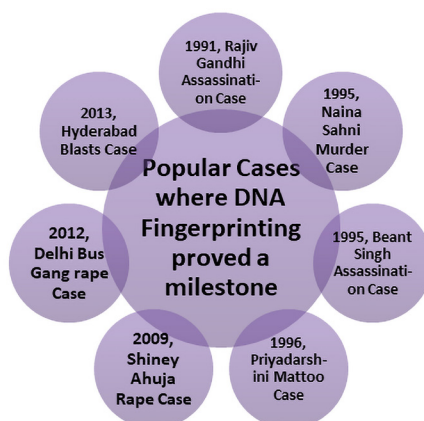
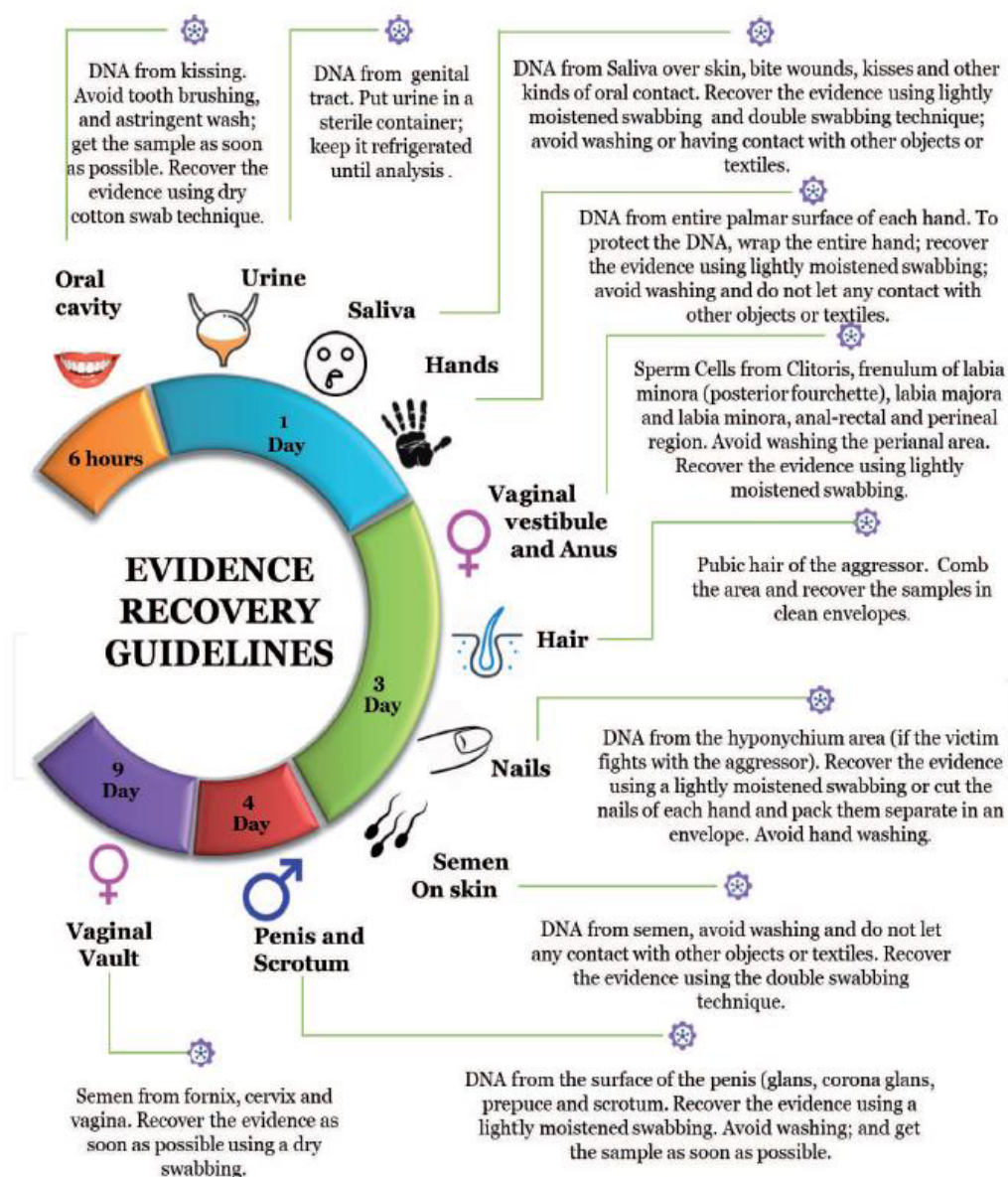


Figure 5: Some popular cases in India where DNA Fingerprinting played a major role Identifying DNA Evidence



Source: *Biological Evidence Analysis in Cases of SA* [4]

Figure 6: Recommended time frame for evidence collection from different anatomical areas according to DNA persistence and its sampling methods in SA cases

Since only a few cells are needed for a useful DNA sample, the list below mentions some areas at the crime scene and on the victim that may contain valuable DNA evidence. Furthermore, DNA does more than just identify the source of the sample; it can place a known individual at a crime scene, in a home, or in a room where the suspect claimed not to have been. The more service providers know about properly identifying, collecting, and preserving DNA evidence, the more powerful a tool it becomes [9].

Table 1: Possible locations of getting DNA evidence and its source [9]

Possible location of DNA Evidence	Source of DNA
Bite mark or any licked area	Saliva
Scrapings from fingernails	Blood, skin cells or hair
Used condom (inside and outside surface area)	Semen and skin cells
Bedsheets, pillows, blankets, or other bed linings	Semen, sweat, hair, saliva or blood
Clothing including undergarments, sanitary napkins	Hair, semen, blood or sweat
Hat or masks	Sweat, skin cells, hair or saliva
A Cigarette butt, toothpick, rim of the bottle, can, or glass	Saliva, epithelial cells
Dental Floss	Semen, skin cells or saliva
Tape/ ligature	Skin cells, saliva or hair

Evidence Preservation

No matter how early the BE has been collected after the assault, if not preserved properly, they are of limited use. Also, with each movement of the victim and passing time, the evidence are very frequently destructed and gets contaminated [10].

1) Avoiding destruction of evidence: A victim needs to know the gravity and potential of BE so that he or she should refrain from various activities before a FME is conducted. It is the duty of the profession to whom the victim had reported the case. Following practices should be strictly avoided [11-16]:

- a) Taking a shower or washing any part of the body, including the mouth, hands, and head hair;
- b) Brushing teeth;
- c) Cleaning or cutting fingernails;
- d) Combing or cutting hair;
- e) Performing vaginal irrigation or douching;

- f) Urinating, defecating or vomiting (and if this is needed, doing it in a clean container with a lid);
- g) Eating, drinking, chewing or smoking;
- h) Running or performing any kind of rigorous physical activities;
- i) Changing, washing or destroying clothing including sanitary napkins worn during the event;
- j) Touching the crime scene.

Also while collecting biological samples; care must be taken in selecting the type of material or container used for sample collection or storage as BE are susceptible to fungal contamination. Complete drying of the container prior to packaging must be assured [17-19].

2) Avoiding contamination: There is a huge probability of genetic material getting contaminated from multiple sources if there is more than one perpetrator; before the medical examination of the victim; during the medical examination; and inside the laboratory [20-22]. Thus to avoid cross-contamination of the samples, examiners must take extra precautions [19, 23, 24]. Some points should be kept in mind while handling biological samples [12, 25]:

- a) Microbial contamination can be avoided by working under aseptic conditions;
- b) All the miscellaneous laboratory items including masks, gloves, gowns, disposables, etc should be used once and then changed for each sample to avoid cross-contamination and also direct contact with the samples must be avoided;
- c) The room where the FME is done should always be cleaned before and after the examination;
- d) Sneezing, coughing, or talking over the samples should be avoided;
- e) Drinking, eating and/or smoking while handling the samples should be strictly prohibited.
- f) Always fresh and untouched cotton swabs should be used.

Moreover, 2-3 sets of samples including the swabs should be collected from the victim, crime scene and perpetrator so that repeat examination can be done in case of cross-contamination of the samples or during further legal proceedings of the case if required.

Conclusion

“Modern Problems Need Modern Solutions”

Because of the unique character of DNA, it is one of the important tools in identifying whether the accused has committed a particular crime especially SO. It is the duty and responsibility of the investigating officer that all the evidence are properly traced with a proper chain of custody and sent for the forensic examination within the prescribed time.

Though DNA testing has not gained much momentum in our country, after the recommendations of the Malimath Committee and Criminal Amendment Act of 2005, it has been now under consideration. The judiciary hence has suggested extending the scope of Section 112 of the Indian Evidence Act. After the incorporation of the Right to Privacy as the fundamental right under Article 21, the court had been subjective regarding the admissibility of DNA evidence on constitutional grounds. Also, Article 20(3) of the Constitution gives the right to a person against self-incrimination. DNA testing is always subject to cross-examination and contradictions.

At the end, the authors want to conclude that there is a need to reconsider such evidence and its admissibility. In the present scenario due to the evolving nature of crimes, the measures taken to prove them guilty should also be updated and evidence like DNA should be given due consideration. Besides this, there should be ‘One Stop Centres’ in different parts of the country on the similar lines of ‘one-stop crisis centre- Gauravi’ in Bhopal, where all the legal formalities, FME, and psychological counselling of the victim can be done to reduce the repeated mental trauma of the victim. In India there are very few labs at present where DNA fingerprinting facility is available, thus government should come up with wider facilities and more such labs that can aid in the investigation and judicial services in SO cases.

With a view to frame legislation for the use and regulation of DNA-based technology in civil and criminal proceedings and identification of missing persons and human remains, the Department of Biotechnology proposed a draft Bill titled “The Use and Regulation of DNA-based Technology in Civil and Criminal Proceedings, Identification of Missing Persons and Human Remains Bill, 2016” The bill, if passed will legalize the setting up of a DNA databank that could be used as a repository of DNA data of criminals easing in correlating the suspect to the crime, victim and crime scene. Though such a concept violates the privacy of a person which is now a fundamental right under article 21 after the famous Justice K.S. Puttaswamy (Retd) vs Union of India case of 2017. But it should be reconsidered in Parliament so that a proper law can be framed on

this. To help in developing the latest technology and research in the field of DNA testing, a Centre of Excellence for DNA is going to be established in Institute of Forensic Science, Lucknow, Uttar Pradesh in collaboration with National Forensic Science University (NFSU), Gandhinagar. This Centre of Excellence to be established in the field of DNA will be the most unique institute in the country. Such positive steps are needed today so that there is more and more research in DNA technology which can aid law enforcement agencies and judiciaries to impart justice to the victim of rape and other heinous crimes.

References

1. Green, S. P. (2016). What Are the SOs? The New Philosophy of Criminal Law. https://law.rutgers.edu/sites/law/files/Flanders%26Hoskins_page%20proofs.61-80.pdf
2. Facts and figures: Ending violence against women. UN WOMEN Accessed on 07-01-2021 <https://www.unwomen.org/en/what-we-do/ending-violence-against-women/facts-and-figures>
3. Pramit Bhattacharya. India is not the most dangerous country for women, data shows. Updated: 28 June 2018, 06:29 PM IST Accessed on 12-01-2021. <https://www.livemint.com/Politics/7PF8NRqgyxoNs2AecfmxVL/India-is-not-the-most-dangerous-country-for-women-data-show.html>
4. González, et al. Biological Evidence Analysis in Cases of SA. Book Chapter Intech Open. DOI: <http://dx.doi.org/10.5772/intechopen.82164>
5. Roa, et al. Identification of male epithelial cells in routine postcoital cervicovaginal smears using fluorescence in situ hybridization. Application in SA and molestation. *Am J Clin Pathol.* 1995 Jul;104(1):32-35.
6. Dziegielewski M, et al. Use of a Y chromosome probe as an aid in the forensic proof of SA. *J For Sci* 2002 May;47 (3): 601.
7. Betz, et al. DYS STR analysis with epithelial cells in a rape case. *For Sci Intl.* 118 (2001) 126-130.
8. Diva Rai. All about DNA Testing in Rape Cases. Updated: 23 June 2020. Accessed on 12.01.2021. <https://blog.ipleaders.in/dna-testing-in-rape-cases-know-its-admissibility/>
9. National Institute of Justice and Office for Victims of Crime. Understanding DNA evidence, A guide for victim service providers. May 2001. <https://nij.ojp.gov/library/publications/understanding-dna-evidence-guide-victim-serviceproviders#:~:text=Understanding%20DNA%20Evidence%3A%20A%20Guide%20for%20>

Victim%20Service%20Providers,NCJ%20Number&text=Given%20the%20importance%20of%20DNA,evidence%20in%20their%20clients'%20cases

10. Magalhães, et al. Biological Evidence Management for DNA Analysis in Cases of SA. *The Scientific World Journal*, vol. 2015, Article ID 365674, 11 pages. <http://dx.doi.org/10.1155/2015/365674>
11. A. Burg, R. Kahn, and K. Welch. DNA testing of SA evidence: the laboratory perspective. *Journal of Forensic Nursing*, vol. 7, no. 3, pp. 145–152, 2011.
12. M. Newton, “The forensic aspects of sexual violence,” *Best Practice and Research: Clinical Obstetrics and Gynaecology*, vol. 27, no. 1, pp. 77–90, 2013.
13. Johnson et al., Use of forensic science in investigating crimes of sexual violence: contrasting its theoretical potential with empirical realities. *Violence Against Women*, vol. 18, no. 2, pp. 193–222, 2012.
14. Benschop et al., Post-coital vaginal sampling with nylon flocked swabs improves DNA typing. *Forensic Science International: Genetics*, vol. 4, no. 2, pp. 115–121, 2010.
15. Magalhaes et al., Forensic procedures: for interview, physical exam and evidence collection in children and young people victims of physical and/or sexual abuse. *Acta Medica Portuguesa*, vol. 24, no. 2, pp. 339–348, 2011.
16. M. Hochmeister and J. Ferrel. SA. The Health Care Response. A Complete Guide to the Forensic Examination and Evidence Collection of the Adult SA Patient, Institute of Legal Medicine, University of Berne, 1999.
17. H. C. Lee and C. Ladd, Preservation and collection of biological evidence. *Croatian Medical Journal*, vol. 42, no. 3, pp. 225– 228, 2001.
18. M. Benecke. Forensic DNA samples—collection and handling. *Encyclopedia of Medical Genomics and Proteomics*, J. Fuchs and M. Podda, Eds., pp. 500–504, CRC Press, New York, NY, USA, 2004.
19. J. M. Butler. *Advanced Topics in Forensic DNA Typing: Methodology*. Academic Press, London, UK, 2012.
20. D. J. Balding and J. Buckleton. Interpreting low template DNA profiles. *Forensic Science International: Genetics*, vol. 4, no. 1, pp. 1–10, 2009.
21. van Oorschot et al., Beware of the possibility of fingerprinting techniques transferring DNA. *Journal of Forensic Sciences*, vol. 50, no. 6, pp. 1417–1422, 2005.

22. Sullivan et al., New developments and challenges in the use of the UK DNA database: addressing the issue of contaminated consumables. *Forensic Science International*, vol. 146, supplement, pp. S175– S176, 2004.
23. J. M. Butler. *Forensic DNA Typing: Biology, Technology, and Genetics of STR Markers*. Academic Press, 2005.
24. J. Anderst. The forensic evidence kit. *Child Abuse and Neglect: Diagnosis, Treatment, and Evidence*. C. Jenny, Ed., pp. 106–111, Elsevier Saunders, St. Louis, Mo, USA, 2011.
25. A. L. Poy and R. A. H. van Oorschot, Trace DNA presence, origin, and transfer within a forensic biology laboratory and its potential effect on casework. *Journal of Forensic Identification*, vol. 56, no. 4, pp. 558–576, 2006.



Combating COVID-19 in Correctional Homes of West Bengal

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Abstract

The world is facing a public health emergency in the form of contagion COVID-19 which has made lakhs of people its victim in a very short span of time. The Correctional Homes, prisons and jails are also affected a lot with the monstrous scourge. Inmates of the correctional homes are one of the most vulnerable section of our population, as they have to reside in close proximity with each other and maintaining social distancing is as good as impossible. In this article, generally I have focused on how to tackle the outbreak of COVID-19 in correctional homes by following the directives received from various agencies. In this article, I have tried to point out the steps taken to combat Covid-19 in prisons such as the circulation of guidelines for awareness generations, personal safety measures, social distancing, sanitation & hygiene, medical attention, decongestion, supervision, activities in correctional homes and subsequent measures. I have explained the consequences of second wave of spread of Covid-19 in relation with convicted prisoners and discussed on statistical data of infection, cure, deceased prisoners and employees. I have mentioned the about the document which required for registration process of vaccination of a person. Finally, with sincere efforts of the officers, men and women posted under the directorate of correctional services of West Bengal, we have been able to combat effectively the Covid-19 pandemic and are continuing our fight sincerely. We intend to get all prisoners and employees of our department vaccinated soon and decongest all Correctional Homes of West Bengal with the help of the judiciary.

Keywords: COVID-19, Correctional Homes, Safety Measures, Medical Attention, Vaccination

Introduction:

A. The world is facing a public health emergency in the form of contagion COVID-19 which has made lakhs of people its victim in a very short span of time. Our Correctional Homes/Prisons/Jails too could not escape from this monstrous scourge.

Inmates of the Correctional Homes are one of the most vulnerable sections of our population, as they have to reside in close proximity with each other and maintaining social distancing is as good as impossible.

B. Category of Correctional Homes of West Bengal with Capacity:

Category		Number		Capacity
Central Correctional Homes	:	08	:	10876

Category		Number		Capacity
District Correctional Homes	:	13	:	4572
Special Correctional Homes	:	04	:	599
Open Correctional Homes	:	04	:	315
Women's Correctional Homes	:	01	:	226
Subsidiary Correctional Homes	:	30	:	2465
Total	:	60	:	19053

C. Prison Population as on 10th June, 2021:

Category		Male	Female	Total
Condemned	:	33	2	35
Convict	:	2635	165	2800
Under Trial	:	19720	1344	21070
Released Inmates	:	364	159	523
Children	:	105	96	201
Total	:	22857	1766	24629 (including 6 Transgender)

Overall Occupancy Rate as on 10th June, 2021:129%

D. To effectively tackle COVID-19 disease outbreak in Correctional Homes, directives were received from various agencies. They are as follows:

- Directives of Hon'ble Supreme Court of India dt. 23.03.2020 and 13.04.2020 in Suo Motu W.P.(Crl) No.01 of 2020.
- Guidelines issued by Ministry of Home Affairs, Government of India vide No.V-17013/17/2020-PR dt.12.03.2020.
- The West Bengal Epidemic Disease, COVID-19 Regulations, 2020.
- Consolidated Guidelines dt. 15.04.2020 of Ministry of Home Affairs, Government of India with regard to measures to be taken for containment of COVID-19.
- COVID-19 Preparedness and Response in places of detention– an Operational Tool box published by United Nations.

Steps taken to combat Covid-19 in Prisons

To combat the menace of Covid-19 the following steps were taken:

I. Circulation of Guidelines for Awareness Generations:

- All Correctional Home Authorities of West Bengal were directed to strictly follow the guidelines circulated by Ministry of Home Affairs, Government

of India vide No. V-17013/17/2020-PR dt.12.03.2020 with regard to precautionary measures in Prisons in view of COVID-19.

- A supplementary guideline was issued by the Directorate of correctional services, West Bengal to all Correctional Home Authorities vide No.136/DG-IG/20 dt.16.03.2020 elaborating preventive measures to be taken by the Correctional Homes.
- All correctional home authorities and other offices under control of the Directorate of correctional services, West Bengal were duly informed about 'The West Bengal Epidemic Disease, COVID-19 Regulations, 2020' with a directive to act accordingly.
- Another order was issued to all concerned vide No. 145/DG-IG/20dt.18.03.2020 to ensure medical screening of all prisoners on very occasion while entering Correctional Home from the courts and to keep under watch all newly admitted prisoners for at least 14 days in an Isolation Ward.
- Posters were provided to all Correctional Homes for awareness of prisoners and employees of correctional homes about the do's and don'ts vis-a-vis COVID-19 without creating any panic.
- Advisory was issued to closely monitor the foreigners recently admitted in correctional homes and the prisoners having history of travelling abroad in the recent past.

II. Personal safety measures, social distancing, sanitation & hygiene:

- Sanitization and spraying disinfectant in correctional homes were done in a regular manner. Sanitization by external agencies like fire brigade and municipal authority also helped.
- Superintendents provided extra soap/liquid soap/sanitizer to prisoners.
- Hand sanitizer was provided to all correctional homes.
- All prisoners and staff were provided with face masks.
- The number of bathing soap issued to the prisoners was doubled.
- Prisoners were advised to maintain social distancing, particularly during interviews.
- Prisoner wards were opened in a segregated manner at different hours to avoid unnecessary gathering.
- Markings were made for visitors to the correctional homes to ensure social distancing.

- Personal Protective Equipment (PPE), hand gloves and face shields were supplied to all correctional homes for use by the medical personnel and paramedical staff in the correctional homes.
- Markings were made on the ground for prisoners to take food maintaining social distancing.

III. Medical Attention:

- Infrared thermometer was issued to all correctional homes.
- Correctional homes undertook constant efforts to ensure medical screening of each prisoner on admission from outside on every occasion.
- Arrangements were made to keep newly admitted prisoners under watch for fourteen (14) days in a separate ward.
- The prisoners having any symptom of cough, cold or fever were shifted to outside government hospital.
- The prisoners, who were in close proximity of the suspected prisoners having symptoms of cough and fever were kept in isolation ward.

IV. Decongestion:

- The Superintendents were advised to move the concerned Ld. courts for ensuring production of inmates through video conferencing to avoid outside contact.
- Follow-up actions were taken in accordance with the recommendations of high-powered committee constituted for recommending cases of certain categories of convicts and UTPs for granting interim parole or bail.
- Special efforts were made for deportation of Bangladeshi prisoners lodged in correctional homes.
- In view of the directives of Hon'ble Supreme Court of India in W.P.(Crl) No.01 of 2020 on 23.04.2020, a high-powered committee was constituted in the State of West Bengal under the chairmanship of executive chairman, state legal services authority, West Bengal. The high-powered committee met on different occasions and made a series of recommendations for interim bail and parole for three months to certain categories of under trial prisoners and convicts.
- 2292 UTPs were granted interim bail through the efforts of district legal services and correctional home authorities. On the other hand, 2770 convicts were granted parole for three months.

V. Co-ordination:

- Superintendents were asked to request the local district/sub-divisional hospital authorities to depute a medical officer to correctional homes having no medical officer to ensure medical screening of the prisoners on their arrival from courts and outside.
- Superintendents coordinated with concerned district police authorities and civil administration for safe transit of prisoners released on interim bail / parole to their residence as per directives of Hon'ble Supreme Court.

VI. Supervision:

- The senior correctional services officials supervised the process and ensured that effective measures were taken by the correctional home authorities.

VII. Activities in Correctional Homes:

- Considering possible health hazards due to outbreak of COVID-19, the work of manufactory units and other vocational and cultural programmes which were regular feature of correctional were kept in abeyance for the time being.
- However, even in such conditions some skilled prisoners were engaged in stitching face masks. Such face masks were tailored in 12 different central and district correctional homes with a total production capacity of 3000 masks per day. These masks were used for internal consumption of inmates and staff of different correctional homes throughout the State. The surplus quantity was supplied to different government agencies like West Bengal police, Kolkata police and CMOH etc. Altogether 30,000 masks were provided to West Bengal police and over 10,000 masks to Kolkata police.
- Production of Personal Protective Equipment (PPE) was initiated at Berhampore central correctional home. The PPEs were prepared as per the standard set by the MSME department, government of West Bengal. The PPEs were sent to Tantuja for quality approval.
- Some prisoners, particularly life convicts showed good will gesture by contributing a considerable amount of money from their wages to West Bengal State emergency Relief Fund. Forty-nine convicts have till date contributed ₹ 2,08,000/-.

VIII.Subsequent Measures:

- In apprehension of possible agitation on the part of prisoners before undertaking the mass transfer of prisoners from one correctional home to another, an exercise to create additional capacity building within the premises of all correctional homes was carried out by utilizing existing infrastructures after their repair for accommodation. As a result of this exercise, additional capacity (temporary) of 2059 prisoners was added to the state correctional homes.
- In respect of few correctional homes, shifting of prisoners from one correctional home to another correctional home was done to manage overcrowding.
- For continuous sanitization of correctional homes, hand spray/ fogging machines were procured for all correctional homes.
- The medical personnel and guarding staff working in close proximity to prisoners and prisoners working in sensitive zones were provided with face shields and mini PPE.
- All prisoners were provided with two face masks.
- COVID-19 tests were undertaken in some correctional homes by the local health authorities of the concerned district/ sub-division.

All correctional homes were provided with the guidelines on home (correctional home) quarantine.

Statistics for the period w.e.f. 23.03.2020 to 31.12.2020

539 prisoners infected 446 prisoners cured and 77 released. 231 employees infected. 227 employees cured.	2770 convicts were granted parole. 2292 UTPs were granted bail. 61923 productions of prisoners were made through video conferencing facility.
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03 prisoners and 02 employees expired

IX. Second wave of spread of Covid-19:

- The correctional homes of West Bengal along with the entire country faced second surge of spread of corona virus in the year 2021 too.
- The high-powered committee of the Hon'ble Calcutta High Court did similar exercise like previous year for the de-congestion of correctional homes as per direction of Hon'ble Supreme Court of India. In its first attempt, 2708 convicted prisoners were granted parole for three months.

Four (4) prisoners and one (1) staff expired due to Covid-19 in 2021 Dilemma

For vaccination of a person anyone of the following documents was required for registration.

- i. Aadhar card
- ii. Driving license
- iii. Health insurance smart card issued under the scheme of ministry of labour
- iv. Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) NREGA Job Card
- v. Official identity cards issued to MPs/MLAs/MLCs
- vi. PAN Card
- vii. Pass book issued by bank/ post office
- viii. Passport
- ix. Pension document
- x. Service identity card issued to employees by central/ state govt./ public limited companies
- xi. Voter ID

However, many of the prisoners did not have any of the above documents. This problem was brought to the notice of the higher authorities of the health department of the government of West Bengal and now this problem has been sorted out and prisoners are getting vaccination without the production of any of the above documents after being identified by the concerned superintendent of the correctional home. Non-release on bail or parole of a large number of under trial prisoners involved in NDPS and POCSO Cases by the Ld. Courts is another headache as they regularly create problem and have to be placated and controlled tactfully.

Conclusion:

With sincere efforts of the officers, men and women posted under the directorate of correctional services of West Bengal, we have been able to combat effectively the Covid-19 pandemic and are continuing our fight sincerely. We intend to get all prisoners and employees of our department vaccinated soon and decongest all correctional homes of West Bengal with the help of the judiciary.

- In the year 2021 till date, 5340 RAT and 913RT-PCR tests were done on prisoners and 367 prisoners found positive.
- 228 RAT and 48 RT-PCR tests were done on employees and 41 employees found positive.
- 3040 employees have been vaccinated out of 3494 in toto.
- 18140 prisoners have been vaccinated out of 24838 in toto.

References:

1. Reports of the superintendents of sixty correctional homes of West Bengal.
2. Official website of honorable Calcutta high court and honorable supreme court of India.



Electricity Theft: Overview, issues, prevention and control in India

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ABSTRACT

This paper gives global overview of the electricity theft cases in general with special emphasis on India in particular. Issues involved in electricity theft, its prevention, steps being taken by the State Governments in this regard and possible steps for ensuring effective control on electricity theft have also been dealt in the paper.

2. Data used:

The data used for the analysis has been taken from 'Crime in India', an annual publication of National Crime Records Bureau of Ministry of Home affairs and 'Report on Performance of Power Utilities' being published annually by Power Finance Company.

3. Electricity theft as per Electricity Act, 2003:

- 3.1 As per the Section 135 of the Electricity Act, 2003, electricity theft occurs when someone: -
- i. taps, makes or causes to be made any connection with overhead, underground or under water lines or cables, or service wires, or service facilities of a licensee or supplier, as the case may be; or
 - ii. tampers a meter, installs or uses a tampered meter, current reversing transformer, loop connection or any other device or method which interferes with accurate or proper registration, calibration or metering of electric current or otherwise results in a manner whereby electricity is stolen or wasted; or
 - iii. damages or destroys an electric meter, apparatus, equipment, or wire or causes or allows any of them to be so damaged or destroyed as to interfere with the proper or accurate metering of electricity; or
 - iv. Uses electricity through a tampered meter; or uses electricity for the purpose other than for which the usage of electricity was authorized.
- 3.2 Punishment for such offence is imprisonment for a term which may extend to three years or with fine or with both.

4. Technical and Non-technical losses in power supply:

4.1 According to World Bank, there are two components to the power losses: technical and non-technical. Technical losses consist mainly of power supply dissipation because of faulty transmission and distribution lines, transformers and measurement systems. There are other losses incurred as a result of actions that are outside the control of the power supply system comprising electricity theft, non-payment by customers, and errors in accounting and recordkeeping. Power theft losses, also referred to as commercial losses, are difficult to estimate but run into huge amounts.

4.2 Non-technical loss (NTL) during transmission of electrical energy is a major problem in developing countries and it has been very difficult for the utility companies to detect and fight the people responsible for theft. Electricity theft forms a major chunk of NTL. These losses affect quality of supply, increase load on the generating station, and affect tariff imposed on genuine customers.

5. Global overview of electricity theft vis-à-vis India:

5.1 Electricity theft is a widespread problem across the world, especially in many emerging market countries. Currently, the top 50 emerging market countries lose \$58.7bn per year due to electricity theft, compared with just \$30.6bn in the rest of the world, including the largest industrialized economies.

5.2 Reducing theft, or “non-technical losses,” in these 50 countries – including the BRICS – is an increasing priority for utilities and regulators. The world loses around US\$89.3 billion annually to electricity theft. The highest losses in India (\$16.2 billion), followed by Brazil (\$10.5 billion) and Russia (\$5.1 billion). India loses more money to theft than any other country in the world. The state of Maharashtra—which includes Mumbai—alone loses \$2.8 billion per year, more than all but eight countries in the world. Nationally, total transmission and distribution losses approach 23% and some states’ losses exceed 50%.

5.3 More than 20% of the electricity generated in India is lost to rampant thefts. Distribution losses are very high and mostly commercial in nature. The three main categories of electricity consumption are agriculture, domestic and industry. Among these, agriculture is widely presumed to be a key contributor to high distribution losses arising from low tariff and extensive theft. Illegal hookings in both the domestic and agriculture categories is rampant and probably constitutes the largest proportion of unaccounted energy. The reasons for this can be traced back to the factors linked to the performance of the utility and the wider socio-political & socio-economic environment.

6. Broad overview of electricity theft in India:

As per the analysis of data published in 'Crime in India' from 2014 to 2020 on the parameters like electricity theft cases registered, cases charge sheeted, cases convicted, cases ended as mistake of factor of Law or civil dispute, cases pending investigation at the end of the year and cases pending trial at the end of the year, the preliminary observations are as below: -

- i. Out of total 36 States/ UTs in India, there are 24 States/UTs [18 States & 6 UTs] with either zero or negligible (below 100) number of cases registered.
- ii. Electricity theft cases have huge regional concentration. The top seven States [Uttar Pradesh, Haryana, West Bengal, Jharkhand, Bihar, Maharashtra and Rajasthan] account for 95-98% of all FIRs. UP and Haryana together contribute more than 80% of the total FIRs in a year.
- iii. Only 8 States [Uttar Pradesh, Haryana, West Bengal, Bihar, Maharashtra, Uttarakhand, Jharkhand and Rajasthan] generally have more than 1,000 cases in a year.
- iv. Most significantly, there are only three States namely Uttar Pradesh, Haryana and West Bengal in which more than 10,000 cases have been registered in a year during 2014 to 2020. These three States together contribute more than 90% of the total number of electricity theft cases registered in India with Uttar Pradesh having highest number of cases followed by Haryana and West Bengal.
- v. In UTs, more than 1,000 cases are registered in Delhi only.

7. Trend of new FIRs (annually) and related parameters of electricity theft in India:

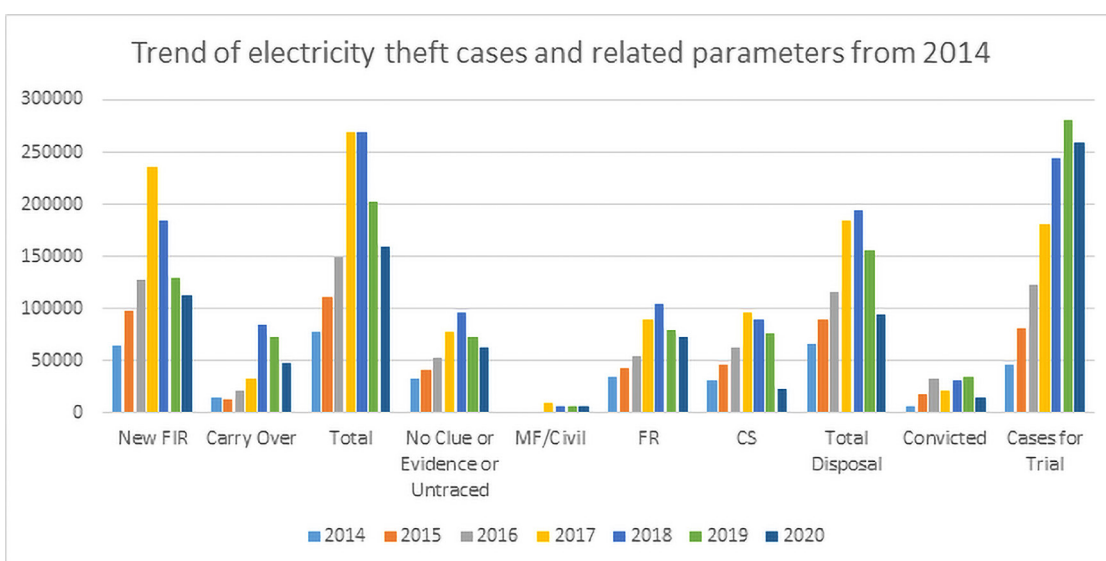
New FIRs cases of electricity theft, carry over cases, 'No clue or evidence or untraced' cases, 'Mistake of Fact (MF)/ Civil Dispute cases, Final Report (FR) cases, Cases Charge-sheeted (CS), total cases disposed by police, cases for trial in the court and cases convicted have been analysed from 2014 to 2020. The data and trend in these parameters is as given below:

Year	New FIR	Carry Over	Total	No Clue or Evidence or Untraced	MF/Civil Dispute
2014	64034	13910	77944	32773	506
2015	97591	12746	110337	41518	721
2016	127910	21727	149637	53178	312
2017	235998	33160	269159	77573	10122
2018	183873	84499	268373	96668	5262
2019	128970	73035	202006	71788	5813
2020	111801	47131	158932	62300	6075

Source: Crime in India, NCRB.

Year	FR	%FR	CS	% CS	Total Disposal	Convicted	Cases for Trial	% Con- victed
2014	34049	43.7	31138	39.9	65196	5242	45305	11.6
2015	42471	38.5	46112	41.8	88602	17248	81764	21.1
2016	53797	36.0	62590	41.8	116392	32538	122255	26.6
2017	88777	33.0	95877	35.6	184659	20310	180489	11.3
2018	104945	39.1	89650	33.4	194601	31438	244161	12.9
2019	78712	39.0	76427	37.8	155164	34138	279997	12.2
2020	71862	45.2	22253	14.0	94509	15132	259150	5.8

Source: Crime in India, NCRB.

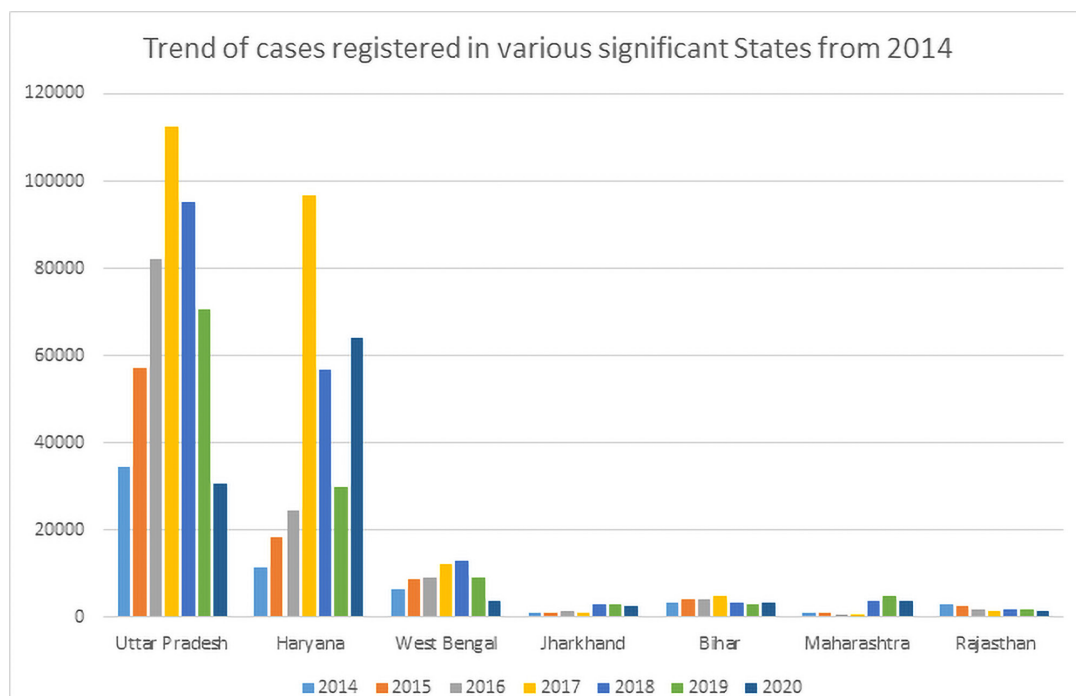


8. Trend of new FIRs (annually) of electricity theft in the significant States:

Analysis of new FIRs cases of electricity theft in various States/UTs has been done with trend in top 7 States from 2014 to 2020. The data and trend is as given below:

	Uttar Pradesh	Haryana	West Bengal	Jharkhand	Bihar	Maharashtra	Rajasthan	Top 7 Share	Gujarat	Karnataka
2014	34534	11517	6581	1035	3347	1140	3196	95.8	161	16
2015	57193	18598	8998	1127	4217	996	2702	96.1	195	8
2016	82343	24517	9250	1356	4326	900	1839	97.4	133	17
2017	112626	96619	12415	1155	5099	929	1524	97.6	103	24
2018	95165	56943	12917	2986	3412	4012	1733	96.4	134	23
2019	70696	30003	9165	2986	3182	5003	2027	95.4	63	14
2020	30657	63995	3893	2723	3268	3743	1567	98.3	51	8

Source: Crime in India, NCRB.



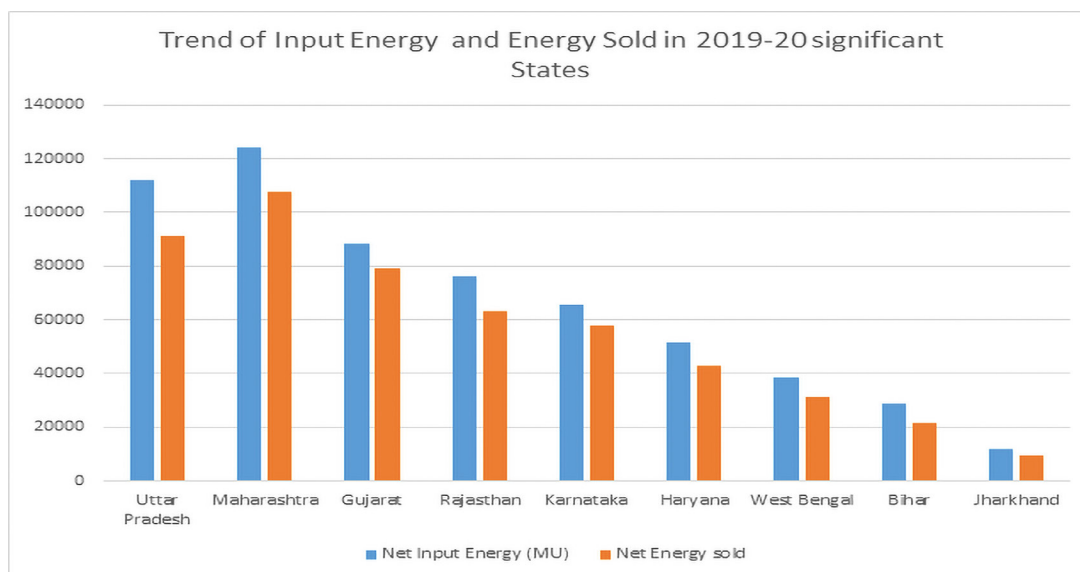
Among the top 7 States, all the States are depicting more or less declining trend in new FIRs in last two-three years except Haryana which has seen a sudden sharp spike in 2020.

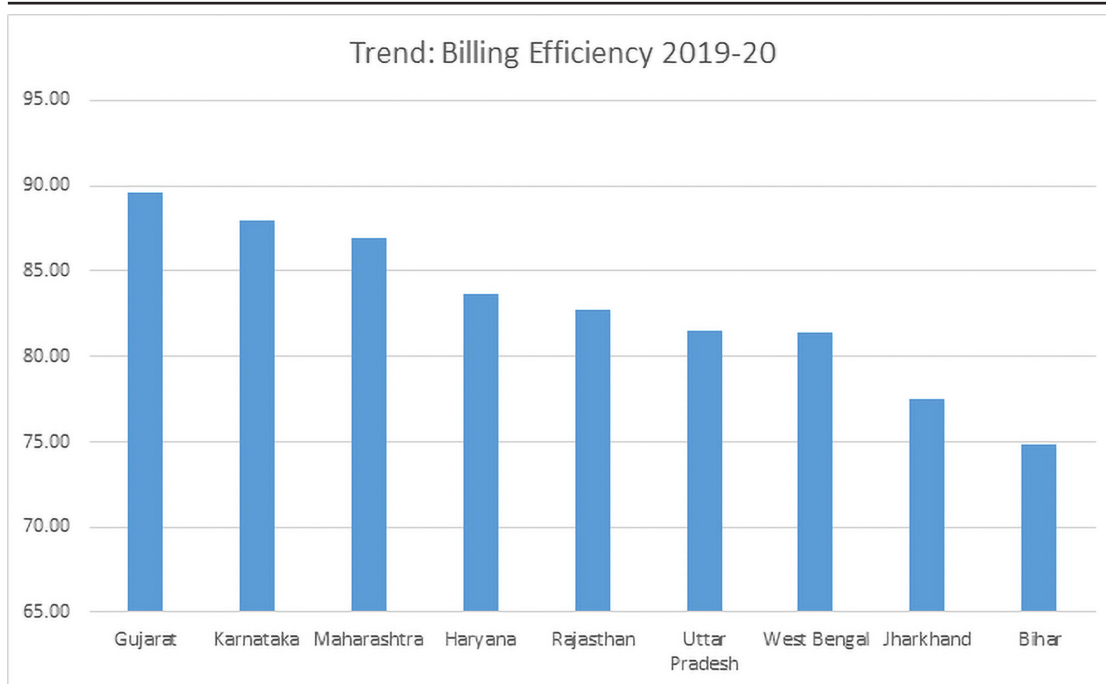
9. Trend of Net Input Energy, Net Energy Sold and Billing Efficiency in the significant States:

From the theft point of view, Billing Efficiency can be a good parameter. Hence, Net Input Energy, Net Energy Sold and Billing Efficiency [Billing Efficiency = (Net Sale of Energy/ Net Input Energy)] is analysed in top 7 States including Gujarat & Karnataka. The data and trend is as given below:

States	2017-18	2018-19	2019-20						
	Net Input Energy (MU)	Net Energy sold	Billing Efficiency	Net Input Energy (MU)	Net Energy sold	Billing Efficiency	Net Input Energy (MU)	Net Energy sold	Billing Efficiency
Uttar Pradesh	111959	88098	78.69	108300	88066	81.32	112225	91486	81.52
Haryana	48733	38236	78.46	49657	40677	81.92	51442	43041	83.67
West Bengal	34915	25598	73.31	36069	2774	77.00	38290	31180	81.43
Jharkhand	11940	9614	80.52	11914	9187	77.11	12013	9315	77.54
Bihar	24197	16779	69.35	26636	18604	69.84	28983	21683	74.81
Maharashtra	118752	101673	85.62	127214	109605	86.16	124117	107866	86.91
Rajasthan	70659	56353	79.75	76096	60326	79.28	76315	63167	82.77
Gujarat	82654	72577	87.81	92531	80093	86.56	88538	79327	89.60
Karnataka	62430	53857	86.27	66952	58337	87.13	65593	57696	87.96

Source: 'Report on Performance of Power Utilities' by PFC.





10. Cases ended as Mistake of Fact or of Law or Civil Dispute:

10.1 The analysis of data related to cases ended as 'Mistake of Fact or of Law or Civil Dispute' during 2014-2020 reveals that there was sudden sharp increase in number of 'cases ended as Mistake of Fact or of Law or Civil Dispute' at All India level in 2017 which though slightly reduced in 2018, 2019 & 2020 but is still significantly very high. The largest contributor in this category is Uttar Pradesh with more than 90% cases.

10.2 States, especially Uttar Pradesh should take steps to improve the quality of FIRs on the basis of hard physical evidences in complaints regarding meter tampering, bypass in meters etc. so as to minimize the number of cases ending as 'Mistake of Fact or of Law or Civil Dispute'.

11. Analysis:

11.1 To analyse whether there is any relation between the number of FIRs registered each year (as published by NCRB) in the above mentioned top 7 States and billing efficiency in these States as published in the 'Report on Performance of Power Utilities' (as published by Power Finance Company), the Pearson's Correlation Coefficient between these two variables has been calculated for 2019 and 2020. For calculating the correlation coefficient, the absolute number of FIRs of electricity theft in these 7 States and billing efficiency in these States has been taken for 2019 and 2020 respectively. The correlation coefficient for 2019 & 2020 is calculated as 0.31 & 0.28 respectively.

11.2 To rule out any possible impact of population size of these States on the annual number of FIRs and its subsequent impact on the correlation coefficient, an alternative approach has been adopted wherein FIRs per lakh of population in these 7 States have first been calculated as below for 2019 and 2020: -

	2019	2020						
	Popula- tion (In lakhs)	New FIRs	FIRs per lakh of popula- tion	Billing Effi- ciency	Popula- tion (In lakhs)	New FIRs	FIRs per lakh of popula- tion	Billing Effi- ciency
U.P.	2259.67	70696	31	81.32	2289.31	30657	13	81.52
Haryana	288.07	30003	104	81.92	292.13	63995	219	83.67
West Bengal	971.09	9165	9	77.00	977.19	3893	4	81.43
Jharkhand	375.81	2986	8	77.11	381.15	2723	7	77.54
Bihar	1201.14	3182	3	69.84	1218.95	3268	3	74.81
Maharashtra	1225.33	5003	4	86.16	1236.76	3743	3	86.91
Rajasthan	776.01	2027	3	79.28	786.09	1567	2	82.77

Source: Report on Performance of Power Utilities', July 2020: Power Finance Corporation.

Subsequently, correlation coefficient between annual FIRs per lakh of population and billing efficiency in these States has been calculated for 2019 and 2020. The correlation coefficient for 2019 & 2020 as per this alternative approach is calculated as 0.31 & 0.27 respectively.

11.3 Interpretation:

Correlation Coefficients calculated as above in both the approaches are more or less same, show a positive correlation between number of cases in States and billing efficiency of utility providers in these States. Despite positive correlation in number of cases and billing efficiency, degree of correlation is moderate and increase in FIRs in electricity theft cases is not increasing the billing efficiency in the same proportion. It indicates that increase in number of FIRs of electricity theft in a State will definitely have a positive impact on the billing efficiency in that State but the impact will be moderate. The possible reasons for the same may be: -

11.1. The offenders involved in big electricity theft (may be in agriculture or commercial activities) are perhaps not getting booked due to which proportionate increase in billing efficiency is not seen.

11.2. There are other significant factors also (both in technical & non-

technical losses) which need to be controlled for effectively increasing the billing efficiency and thereby reducing the power loss either through theft or otherwise.

12. Initiatives/ Steps taken by the State Governments for controlling the power theft:

12.1 In the Vigilance unit of UP Power Corporation Ltd, 33 enforcement teams were engaged for controlling the electricity theft. For ensuring the effective control on electricity theft, in 2017 UP Government decided to increase number of these enforcement teams to 88. In 2018, UP Government further decided to establish one anti power theft police station in each District (total 75 such police stations). In these 75 Anti power theft Police stations, it is ensured that all the cases of electricity power theft under the sections 135,136,137,138,139,140,141 and 150 of Electricity Act-2003 of a district are registered and investigated in one such police station of the concerned District.

12.2 Majority of States are going for replacement of open/ live wires by underground covered electricity cables for transmission of electricity to households. Additionally, smart meters are also being installed in majority of States for detecting and controlling the unusual pattern in electricity consumption. These steps are definitely helping in reduction of power theft but there is still a lot to be done for substantially curbing the power theft in the country and saving the public money.

12.3 A Report of the Task Force Private Participation in Power Distribution by Secretariat for Infrastructure Planning Commission, Government of India, July 2012 wherein the prospects of private participation in power sector have been analyzed. The report suggests that given the importance of the power sector to the economy, the paucity of budgetary resources and the deteriorating financial health of the distribution companies, it is important to restore order in the power distribution sector. Private participation in the power distribution sector should be pursued in order to attract investment and to harness the efficiencies of the private sector. In this regard, the Task Force endorsed both the models, i.e., the Public Private Partnership Model and the Franchisee Model, leaving it to the States to choose a model that they think is more suited to their needs.

13. Conclusions:

13.1 Electricity theft is not a new but a complex phenomenon with many facets. Theft of power is common in many countries and a considerable amount of energy is stolen every year from the electric low voltage grid. An electric utility

low voltage grid can never be one hundred percent secure from theft and fraud due to the number of distributed end points which sit largely unmonitored in public accessible locations.

13.2 Non-technical losses have plagued utilities since their inception and are now a global problem for electric providers having huge financial implications. Non-technical losses are derived from socioeconomic, political, environmental and technical roots, but the end solution is generally sought solely through technical measures. Non-technical losses drive up costs for the consumers and cut into the utility's revenue and over all bottom line.

13.3 The nexus between customers and employees of the Power utilities with intent of corruption also gives rise to electricity theft in socioeconomically depressed countries like India.

13.4 The existing policies focusing on harsher punishment models, more robust enforcement, or technology-based solutions have definitely given result in curbing electricity theft but not to the desired level.

13.5 Social acceptability is higher for theft offenders with less well-off income and power supply quality contexts. However, raising punishment levels to deter theft may not necessarily be socially desired.

13.6 Overall, the desired punishment levels for power theft are low. While there exists a sense of social reprimand for stealing power, desired punishment is nuanced and context-dependent.

13.7 The more well-off an individual is and with better supply quality, the less acceptable it is to be committing theft, and vice versa.

13.8 Greater literacy and greater income are closely associated with lesser power theft.

14. Way Forward:

14.1 Addressing the social norms at play and considering a norm shift could significantly impact accountability and the provision of public services.

14.2 Social norm adjustments and participatory approaches could contribute to curbing theft and improving service provision.

14.3 Power Utility Companies should go for smart grid infrastructure to combat this theft, boost reliability and improve the financial sustainability of utilities.

14.4 Increasing the share of private installed capacity can also make



significant impact.

14.5 Electricity Act, 2003 could be amended to with an intention to a policy shift from FIRs, arrests & convictions to substantial fines deposited to Utilities. Provisions for inclusion of private partnership, as suggested by the Report of Planning Commission, may also be included in the law.

References:

1. Katiyar, EPW, 40 (2005)-Political economy of electricity theft in Rural Areas.
2. Jason Wong, World Development 140, 2021-Perceptione re electricity theft, role of social acceptability regarding electricity theft in UP.
3. Determinants of electricity theft-Ener Pol; 93 (2016) V. Gaur and E. Gupta-Corruption, Tax/GDP, collection efficiency, private capacity, literacy and income.
4. Annual Emerging Markets Smart Grid: Outlook 2015 study.
5. 'Crime in India' 2014 to 2020, Annual publication of NCRB.
6. 'Report on Performance of Power Utilities' 2018 to 2020, being published annually by Power Finance Corporation.



Project Management of CCTNS (Crime and Criminal Tracking Network & Systems)

(A system connecting all police stations in the country
through an online platform)

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Abstract

Crime and Criminal Tracking Network & Systems (CCTNS) is a Mission Mode Project of the Government of India, to create a comprehensive and integrated IT system for enhancing the effectiveness of policing at the Police Stations and Higher Offices with an outlay of `2000 Crores. Project management of e-governance projects of this magnitude requires an out of box approach challenging traditional methods. Typical challenges were the involvement of multiple stakeholders and varied levels of IT.

Introduction

In an effort to modernize the police force, the Ministry of Home Affairs conceptualized the Crime and Criminals Tracking Network & Systems (CCTNS) project as a Mission Mode Project under the National e-Governance Plan (NeGP) during the year 2009. It focused on the modernization of the Police to improve outcomes in the area of crime investigation, criminal detection, information gathering and its dissemination across various police organizations and units across the country and to enhance services to citizens.

CCTNS was further expected to help in the deployment of the Interoperable Criminal Justice System (ICJS), which aimed at integrating police data under CCTNS with prisons data, prosecution, forensics and courts.

CCTNS was conceptualized to automate the police working. It facilitates the entry of data related to crime reported at police stations including details of the scene of crime, accused involved, property seized, and final report filed in the court. It also provided a search facility for persons and property, which makes the investigation process easy and faster. NCRB as one of the main stakeholders was entrusted with monitoring and coordination of this project at the National level.

Scope of the CCTNS

It was envisaged to achieve the following outcomes from the project:

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- a. It aims to cover more than 16000 police stations and 6000 higher offices.
- b. Make the Police functioning citizen-friendly, transparent, accountable, effective and efficient by automating the processes and functions at the level of the Police Stations and other police offices at various levels.
- c. Improving delivery of citizen-centric services through effective usage of Information & Communication Technology (ICT).
- d. Provide the Investigating Officers with the tools, technologies and information to facilitate faster and more accurate investigation of crime and apprehension of criminals.
- e. Improve the Police functioning in various other areas such as Law & Order, Traffic Management and resource management etc.
- f. Facilitate the collection, storage, retrieval, analysis, transfer and sharing of data and Information among Police Stations, Districts, State headquarters and other organizations/agencies, including those at the Government of India level.
- g. Enabling and assisting the senior Police Officers in better monitoring and management of the Police Force.
- h. Keep track of the progress of the crime and criminal investigation and prosecution Cases, including the progress of cases in the Courts.
- i. Reduction in manual record keeping.

Challenges in CCTNS Project Implementation

Implementation of such a big project was not an easy task. There were a lot of unprecedented challenges in the project such as:

- 1. Project spreading over 20000+ locations
 - a. Very huge project and was initially proposed to cover more than 16000 Police Stations and 6000 Higher Offices.
 - b. States/UTs have different geographical, financial and governance conditions.
- 2. Multiple Stakeholders
 - a. The project had many stakeholders like States/UTs, MHA, NCRB, NIC, BSNL, System Integrator, Software Development Agency, Central Project Management Unit, State Project Management Unit etc.
 - b. All Stakeholders were not under the direct control of NCRB.
 - c. Difficulty in coordination between different stakeholders.

3. Remote Locations

Many Police Stations in the States/UTs were located at the remote location where Road/Rail Transport was not available. This led to issues in the deployment of Hardware Infrastructure and providing network connectivity

4. Different level of readiness of States/UTs

- a. Advanced States (like Andhra Pradesh, Telangana, Karnataka, Tamil Nadu and Gujarat) were already using their software for police functioning and they were quite ahead in implementation.
- b. Some states were not able to proceed due to their managerial and financial conditions.
- c. Some states/UTs had ready Infrastructure while some states did not have proper building/infrastructure to deploy the CCTNS Hardware.

5. Difference in performance of System Integrators

The performance of System Integrators selected by different States/UTs was not same. Some of the System Integrators could not perform their tasks and adhere to contractual conditions. Due to poor performances in many states, the contract with System Integrators was terminated and the selection of new System Integrator caused delay in the implementation of the project.

6. Capacity Building

There was a need of change management and basic IT training which was a herculean task and police personnel reaching to age of retirement were not willing to learn new IT based system.

7. Data Digitization

It was decided to digitize 10 years of legacy data but many old physical records were not available to digitize and many were not having mandatory information like age, address etc. as well as some old records have junk/irrelevant information

Evolution of Evaluation and Monitoring Mechanism

CCTNS had adopted a “Bundle of service” approach rather than procuring solution components (such as hardware, networking equipment, application, capacity building services, etc.) separately. All activities under these basic components were being monitored to streamline the progress of the Project. Therefore, Project management played an important role in the implementation of the project. NCRB evolved mechanisms and strategies to monitor the project and revised them from time to time as per the requirement.

Daily/Monthly Dashboard

To ascertain progress and regular monitoring of the CCTNS project, NCRB had started collection of information periodically in year 2012 on the infrastructure parameters like readiness of infrastructure, Hardware delivery and commissioning, capacity building, Network connectivity, establishment of State Data Centre (SDC) and Digitization. Based on information received from the States/UTs, a Daily and Monthly Dashboard was used to prepare to show the status of basic project implementation activities. This Monthly Dashboard was used to review the status of CCNTS in various meetings of MHA/NCRB. As the project implementation moved along, these parameters changed with the changing requirements.

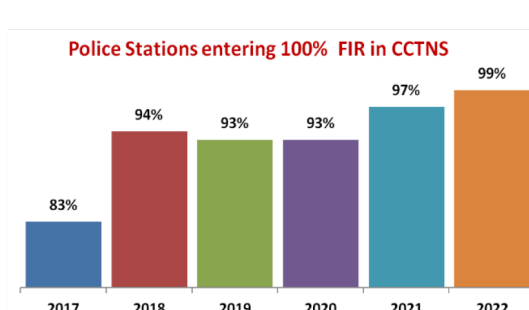
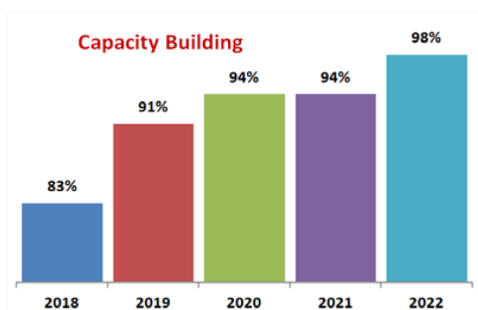
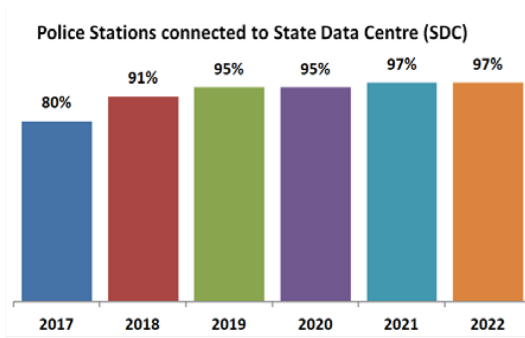
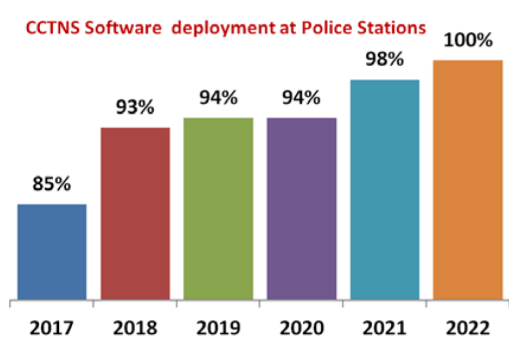
PRAGATI Dashboard

In December 2015, CCTNS Project was included in PRAGATI Review. NCRB converted its dashboard into a more extensive PRAGATI Dashboard for monitoring the project progress across all States and UTs. The dashboard is prepared monthly. Initially, the PRAGATI Dashboard was containing parameters based on Hardware/Software Deployment and Network connectivity (like Total No. of Police Stations in States/UTs, No. of PSs where CCTNS Software is deployed, Data Migration done, No. of PSs having Network Connectivity, NKN Connectivity established for SDC-NDC, States/UTs Data Replication started at NDC), No. of Police Stations entering IIF 1 to 5 Forms in CCTNS and No. of IIF 1 to 5 Forms entered in CCTNS.

Revisions in PRAGATI Dashboard

Later in 2016, many revisions have been made to the monitoring mechanism for developing PRAGATI Dashboard. Some of these are as follows:

- Major activities/parameters were given weightage based on the percentage of completion of the activity. It enabled us to identify the gaps, which were followed up with the States/UTs.
- An Activity Wise Sheet was conceptualized which shows the overall percentage as well as the percentage of all major activities, which created a healthy competition among states to eliminate gaps and score higher and higher.
- The percentage based coloring scheme was defined to improve visualization related to the project's progress.
- A map of India was included in the Dashboard while showing the performance of States/UTs through a defined color scheme.



Total No. of Police Stations : 16453

No. of States/UTs	Implementation %
19	90 + %
8	80-89%
3	70-79%
3	60-69%
1	50-59%
2	40-49%
0	30-39%
0	20-29%
0	10-19%
0	0-9%

- New parameters were added in the PRAGATI Dashboard from time to time and the weightage of each parameter was defined on the basis of the importance of the activity. The weightage of the progress in each activity was calculated based on the percentage of their progress. The performance was divided into 10 categories and the color scheme according to their categories was developed. It provided better visualization of the progress and enabled the concerned authorities to take action accordingly.
- Later, it was felt that despite the Hardware and software deployment at the PS level, use of CCTNS at the Police Station level was very low. To boost up the speed of Project Implementation and to increase the use of the Application, some more parameters related to use of CCTNS software were added in the Dashboard. The weightages of the activities/parameters were also revised to focus on the low performing activities. As a result, entry of IIF forms and use of CCTNS increased accordingly.
- To facilitate the Citizens with the benefits of CCTNS, monitoring of Citizen Services was included in PRAGATI Dashboard. It helps not only in starting of Citizen Services through CCTNS but also to increase the use of Citizen Services. As of now, more than 10 Crore requests for various citizen services have been received through Citizen Portal.
- Based on the analysis of the PRAGATI Dashboard, NCRB started organizing CCTNS Review meetings with Nodal Officers of States/UTs to discuss the status of the project, to make a time plan to fill the gaps in implementation. States/UTs were also instructed to convene regular SAC (State Apex Committee Meeting chaired by Chief Secretary) and SEC (State Empowered Committee Meeting chaired by DGP) Meetings to review the progress and to improve the implementation speed.
- States/UTs were divided in to the following four categories:
 - a. Advance States (Non CAS using State/UTs)
 - b. Major States including Delhi
 - c. Hilly States
 - d. Union Territories excluding Delhi

](CAS using States UTs)

It made the Ranking more justified among States/UTs as they are compared with peer States/UTs. States/UTs were motivated to increase their ranking in PRAGATI Dashboard.

Conference on Good Practices in CCTNS and ICJS

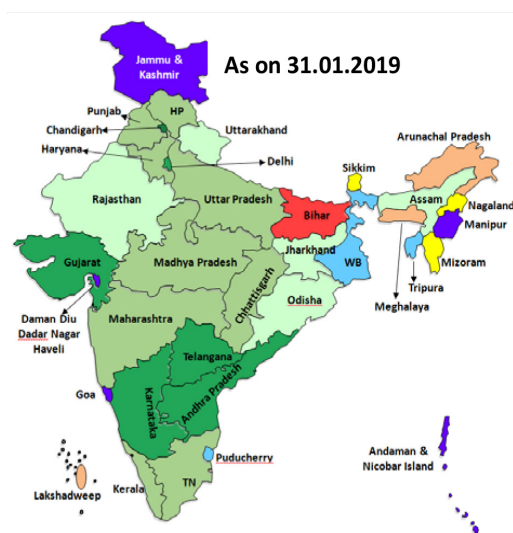
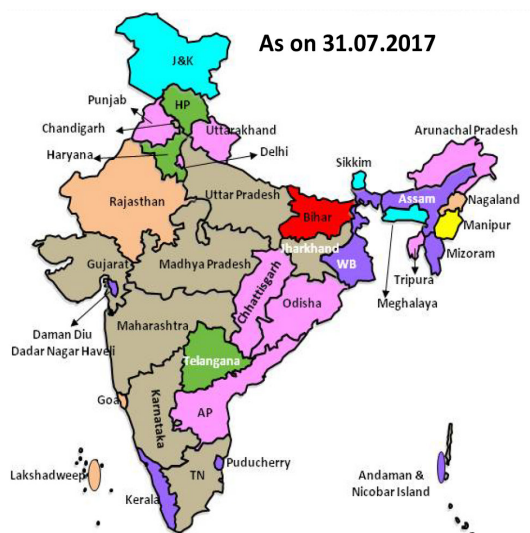
To boost the use of CCTNS and to motivate the States/UTs for using CCTNS, Conferences on Good Practices in CCTNS were initiated. At this conference, the best performing States/UTs and individuals are awarded. Recently, NCRB has conducted two days conference on Good Practices in CCTNS & ICJS in December 2021. In this conference, total of 4 Awards to best performing States/UTs in CCTNS, 12 Awards to the top 3 States/UTs in ICJS and 58 Individual Awards for their significant contribution

to CCTNS were given.

- **Compendium of CCTNS/ICJS Good Practices and Success Stories**

During the implementation of the project, many states and UTs have adopted innovative practices and ideas. These practices and ideas have been compiled in a form of a compendium which is released by NCRB for reference for other States/UTs.

Impact



all units (16453 PSs & 7760 HOs). Latest PRAGATI Dashboard as of 01st April 2022 consists of total of 64 pages including India Map, Activity wise sheet, graphs on major activities, ICJS Dashboard, 27 compiled input forms and the latest Fund utilization status.

Conclusion

Monitoring any project plays an important role in its implementation. PRAGATI Dashboard has been made a basic tool to major/monitor the performance of the CCTNS Project. It helps not only to improve the project's progress but also to achieve most of the Project's targets. PRAGATI Dashboard is being used by all States/UTs/MHA/NCRB/other stakeholders to check the Status of the CCTNS Project Implementation. Various Parliament Questions related to CCTNS Project Implementation are replied using the PRAGATI Dashboard. It shows the progress through visual impression (using color scheme) through which progress in the area can easily be observed and it has a high recollect value. It provides a single platform for all Stakeholders to check or access their progress/status in any activity. It has created a healthy competition among all States/UTs which motivated States/UTs to go up in the ranking list of the PRAGATI Dashboard. For improving implementation of CCTNS Project, Top ranking States/UTs are awarded in Good Practices Conference organized by NCRB.



Anticipatory Bail for Children in Conflict with law: Issues & Challenges

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Abstract

The article identifies the dilemma which exists today with regard to the maintainability of an application for anticipatory bail in case when the applicant is a juvenile. For this just-mentioned issue, there are simultaneous contrary opinions from the various High Courts. Some of them have answered in the affirmative and others disallowing the application seeking anticipatory bail of juveniles. This article analyses the relevant provisions and the judgments involving contrary opinions in addition to the logic followed by the various High Courts. There are two contrasting opinions; one allowing the application for anticipatory bail in case of juveniles and thereby giving him/her the opportunity which the adults have. The other, denying the applicability of general law when there is in existence a special law for juveniles, though the special legislation is silent on the matter. It also highlights certain related rules of interpretation which are applicable to this issue. Finally, the article advocates for the maintainability of application for anticipatory bail providing reasons in its favour. The research method of the article is primarily doctrinal based and an analysis of legal principles, legislative provisions involved, judicial pronouncements, etc.

Keywords: Anticipatory bail; juvenile; special law; general law; legislative intent.

Background

On 20th January, 2022, Calcutta High Court requested the Hon'ble Chief Justice to constitute a larger Bench to decide as to whether or not an application for anticipatory bail under section 438 of the Code of Criminal Procedure, 1973 at the instance of a minor/juvenile is maintainable, in view of the fact that there is divergence of opinion between coordinate benches of this Court in that regard.³ The matter pertained to an application for anticipatory bail on behalf of a minor. The issue is controversial because there are inconsistencies among a general law and a special law with regard to such matter. Before dealing with anticipatory bail in context of juveniles, a look at the concept of anticipatory bail is required. Anticipatory bail refers to a kind of bail which is sought by an applicant who has

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3 Suhana Khatun & Ors. v. State of West Bengal CRM No. 2739 of 2021.

an apprehension that he/she may be arrested in near future for the commission of a non-bailable offence. Statutory provision which provides for this kind of bail is section 438 of the Code of Criminal Procedure, 1973. An anticipatory bail is different from regular bail as in case of regular bail the accused has been in custody and already been produced and have entered into the criminal justice system. The need for granting anticipatory bail arises mainly because there can be false cases; where sometimes influential persons try to implicate their rivals in false cases for the purpose of disgracing them or for other purposes by getting their rivals detained in jail for some days. There can be cases where there is no likelihood of absconding of the accused person or the applicant. Reasonable grounds exist for holding that the applicant is not likely to abscond or otherwise misuse his liberty while on bail. In the above mentioned situations, there seems no justification to require him first to submit to custody, remain in prison for some days and then apply for bail. The very purpose for the provisions relating to anticipatory bail is to ensure that no person is confined in any way until and unless held guilty. As the concept of anticipatory bail has been dealt under the Code of Criminal Procedure, 1973 is a general law and will not be applicable to persons who are dealt according to special laws viz., children in conflict with law as for them, there exists the Juvenile Justice (Care and Protection of Children) Act, 2015. At the same time, there is absence of any provision in the special legislation that provides for the anticipatory bail for children.

Research Objectives

- The objectives of writing this research article are:
- To highlight the existing legislative position with regard to anticipatory bail as provided in the Code of Criminal Procedure, 1973.
- To identify the distinction with regard to provision of anticipatory bail for adults and juveniles.
- To analyse the contrary views taken by various high courts in this regard.
- To suggest the view which seems more appropriate with regard to the issue at present?

Legal Material and Method

This article is primarily based on doctrinal method of research in which analysis of various judgments have been done along with relevant legal provisions. The reasoning for contrary opinions has been discussed in order to appreciate the decisions. The article has also referred the secondary data available on internet including various case comments available on this issue and newspaper articles related to it.

Law on Anticipatory Bail

Section 438 of the Code of Criminal Procedure, 1973 provides for anticipatory bail. It provides that any person having reason to believe that he may be arrested on accusation of committing a non-bailable offence, may apply to the High Court or Sessions Court for a direction that in the event of arrest, he shall be released on bail. It is only on arrest that the order granting anticipatory bail becomes operative.⁴ While pondering over the question of granting bail, there are various factors which the Court considers. The factors include, nature and gravity of accusation, the antecedents of the applicant including the fact as to whether he has previously undergone imprisonment on conviction by a court in respect of any cognizable offence, the possibility of the applicant to flee from justice, where the accusation has been made with the object of injuring or humiliating the applicant by having him arrested etc. The Court while granting anticipatory bail can impose certain conditions to be followed by the applicant while on bail. These conditions have been mentioned under section 438(2) and includes such as the applicant has to make himself available for interrogation by a police officer as directed by the Court or as required by the police officer, he/she should not leave the country without the previous permissions of the Court, he/she should submit local residential address, native address and contact number to the concerned police station, he/she should not make any inducement, threat, promise etc. to any person acquainted with the facts of the case. The before-mentioned conditions are not exhaustive and there can be other conditions imposed by the Court to strike a balance between individual's right to personal freedom and the investigational rights of the police.

Identifying the dilemma

Although the position of law with regard to anticipatory bail, in general, is a settled one, the position regarding its applicability in case of juveniles or children in conflict with the law is in a state of flux. Due to the absence of an authoritative pronouncement by the Supreme Court on the issue, different benches of various High Courts have taken conflicting views and supported them with their own reasoning. The dilemma is existing because though the general law i.e., The Code of Criminal Procedure, 1973 provides for the provision for anticipatory bail, the issue pertains to the applicability of that provision in case of juveniles. As for the fact, juveniles are to be dealt according to the special legislation i.e., the Juvenile Justice (Care and Protection of Children) Act, 2015. This special legislation which applies to children alleged to be in conflict with law is silent on the matter of anticipatory bail. The provision which deals with bail of a child

⁴ Balchand Jain v. State of M.P., (1976) 4 SCC 572

alleged to be in conflict with law is section 12 of the Juvenile Justice (Care and Protection of Children) Act, 2015. Section 12 provides for a non-obstante clause which excludes the application of the provisions of the CrPC whenever a child is apprehended or detained by the police. Also, the special law does not use the word, 'arrest' with regard to a child in conflict with the law. Instead, it uses the term 'apprehension'. Now, the question that arises and needs to be answered is whether a child's application seeking anticipatory bail is maintainable or not.

Discussion

This part of the article deals with the two contrasting views taken by various High Courts with regard to the matter in issue. The views have been categorised into two approaches viz., literal and purposive.

Literal Approach

When the literal approach is considered the arguments which are advanced refer to one of the principles of interpretation i.e., *generalia specialibus non derogant*; general law does not prevail over special law. Applying this would mean, that the Juvenile Justice (Care and Protection of Children) Act, 2015 would prevail over the Code of Criminal Procedure, 1973. Further, as Section 12 of the Juvenile Justice (Care and Protection of Children) Act, 2015 provides for a non-obstante clause which excludes the application of the provisions of the Code, Section 438 would not be applicable in case of juveniles.⁵ Also, the Chhattisgarh High Court held that the grant of bail to a juvenile is required to be dealt with under section 12 of the Juvenile Justice (Care and Protection of Children) Act, 2015 and not under sections 437 or 439 of the Code. This conclusion was reached due to non-obstante clause mentioned in the Juvenile Justice (Care and Protection of Children) Act, 2015.⁶ Also, there have been decisions where it was held that the Juvenile Justice Board does not possess the power which is equivalent to section 438 of the CrPC.⁷

In addition to the presence of a non-obstante clause, there is an absence of an enabling provision which empowers either the Juvenile Justice Board or High Court/Sessions Court. It has been argued that without an enabling provision, the relief of anticipatory bail should not be granted. There is no provision either in the Juvenile Justice (Care and Protection of Children) Act, 2015 or in the Code which enables the juvenile to move an application for anticipatory bail either before the Court of Sessions or High Court or even before the JJ Board.⁸

5 In Re: Krishna Garai Cal. 2016.

6 Tejram Nagrachi v. State of Chhattisgarh 2019 SCC OnLine Chh 24

7 Sandeep Singh Tomar v. State of M.P. M.Cr.C. No.10345 of 2019, Preetam Pathak v. State of Chhattisgarh M.Cr.C.(A)No.1104 of 2014.

8 Kamlesh Gurjar v. The State of Madhya Pradesh M.Cr.C. No.10345 of 2019.

Another argument against the application of anticipatory bail is that the provisions of the Juvenile Justice (Care and Protection of Children) Act, 2015 do not envisage arrest of the children in conflict with the law. The Parliament has deliberately used the term apprehend instead of arrest. As the prerequisite of apprehension of 'arrest' under Section 438 does not exist, the power under the provision cannot be exercised by a High Court or a Court of Session in granting anticipatory bail to a juvenile. Justice Nagamuthu of the Madras High Court argued in a similar way and noted that the intention of the Parliament was not to empower a police officer to arrest a child in conflict with the law rather only to apprehend them under the Juvenile Justice (Care and Protection of Children) Act, 2015.⁹ The Court held that there cannot be any fear of arrest in the case of a child in conflict with the law and this legal position obviates the need for them to seek anticipatory bail.

After pondering over the above-mentioned arguments, the literal approach of the relevant provisions leads to such conclusion that the application for anticipatory bail in case of juveniles is not maintainable.

Purposive Approach

With regard to the interpretation of the provisions of the Juvenile Justice (Care and Protection of Children) Act, 2015 generally and Section 12 of the Act particularly, it is an established principle in law that in case of conflict or inconsistency between general legislation and special legislation, the provisions of special law prevail to the extent of that inconsistency. However, where there is no inconsistency and the general legislation provides something which is not prescribed by the special legislation, the provisions of general legislation would prevail. The Allahabad HC observed that the provisions of the Code may apply and operate in areas where the Juvenile Justice (Care and Protection of Children) Act, 2015 is either silent or constructs no special or distinct measure. It was held that the non-obstante clause in Section 12 is only indicative of the JJ Board being conferred the power to grant bail notwithstanding any restraint or fetter that may be found in that regard in the Code.¹⁰ Moreover, a special court in Delhi held that there is no exclusion of granting the benefit of anticipatory bail to a child in conflict with the law as the said benefit under section 438 of the Code is available to "any person". The Court while appreciating the objectives and the purpose of the Juvenile Justice (Care and Protection of Children) Act, 2015 emphasized that a court has a duty to see that the rights of a child in conflict with law are not in any way impaired. It was held that a juvenile apprehending arrest

9 K. Vignesh v. State 2017 SCC OnLine Mad 28442.

10 Shahaab Ali and Ors. v. State of U.P. 2020 CriLJ 4479 of Hon'ble Allahabad High Court.

will be entitled to seek the discretionary relief of pre-arrest bail envisaged under section 438 of the Code as the provision takes within its ambit “any person” to seek such relief.

The Allahabad High Court in Shahaab Ali case also clarified that the words “arrest” and “apprehend” can possibly be used as substitutes for each other and convey an identical meaning. In fact, the word ‘arrest’ has been replaced by the word ‘apprehension’ for the reason that there is a principle of use of non-stigmatizing semantics. Any term which might have a derogatory or negative impression has not been used in the special legislation made for children. It was held that Section 438 of the Code will apply at the stage before the registration of First Information Report (‘FIR’) and it is within this narrow confine that the right to invoke the jurisdiction of the Court of Sessions or the High Court to grant anticipatory bail to juvenile must be recognized to exist and preserved. After the registration of an FIR, Section 438 stands impliedly excluded and only Section 10 (which provides for the apprehension of juvenile) and Section 12 of the Juvenile Justice (Care and Protection of Children) Act, 2015 would operate.

Consequently, the purposive approach suggests maintainability of the application seeking anticipatory bail in case of juveniles.

Conclusion and Suggestion

This article has advocated the idea that juveniles must also be allowed to apply before Sessions Court or High Court in order to seek anticipatory bail as an adult can, irrespective of the fact that there is absence of any enabling provision in the special law. In fact, the special law is silent about the anticipatory bail and therefore, there is no inconsistency between the two; general and special law. Also, based on the arguments discussed in the previous section, the purposive approach seems more appropriate.

The purpose of Section 438 was to recognize the importance of personal liberty and freedom in a free and democratic country. The legislative intent and the age-old principle that an individual is presumed to be innocent till he is found guilty by the court is reflected in the anticipatory bail.

Both the above mentioned points attain more relevance in context to juveniles. Suppose, there is a person who has been caught red-handed by police committing a non-bailable offence and during investigation, he reveals names of two persons; one X and the other Y. X is a major and Y is a minor. Now, X files an application before the High Court seeking anticipatory bail. Y does not file such application as there is no provision enabling him to apply

for anticipatory bail. This illustration points to the direction that the law of anticipatory bail is not uniform when compared to that applicable to adults and juveniles. In fact, there exists no legal provision with regard to anticipatory bail of juveniles. And, adults in the system are getting the opportunity of availing the benefit of this anticipatory bail, but a child cannot. As the Juvenile Justice (Care and Protection of Children) Act, 2015 is a beneficial and remedial legislation; such a rights-based approach is in the welfare of the children in conflict with the law. The Courts emphasized the duty of a Court to ensure that the rights of juveniles do not get violated. Excluding a complete category of persons from the ambit of such a relief when the provision expressly allows “all persons”, has far-reaching consequences. It would violate the spirit of the Juvenile Justice (Care and Protection of Children) Act, 2015 and would defeat the purpose for which provision of anticipatory bail was enacted.

The Supreme Court should intervene and conclusively settle the position of law as it greatly impacts the liberty and well-being of such children. The Juvenile Justice (Care and Protection of Children) Act, 2015 would not be able to achieve its full potential if children do not get the relief of anticipatory bail due to technicalities in the procedural law.



Haryana 112: Emergency Response is just a call away!

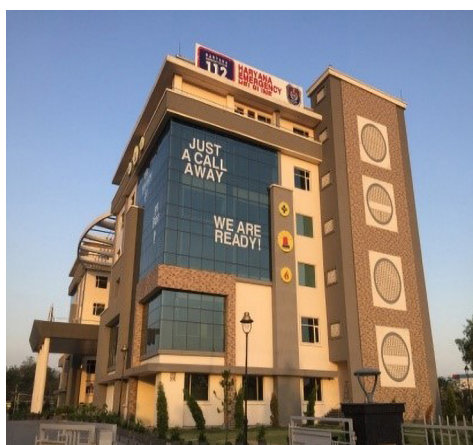
Dr. Arshinder Singh Chawla, IPS

Addl. Director General of Police (Admin & Telecom-IT) Haryana

Haryana 112 – Emergency Response Support System (ERSS) is a flagship project of the Government of Haryana, initiated to provide prompt and integrated emergency response to the general public in distress.

Haryana 112 project was conceptualised with a mission to provide high-end safety to the general public of the State. The project is efficaciously benefitting more than 25 million citizens of the state by leveraging advanced technologies and global best practices of emergency response. The project is designed to function within an average response time of 15 minutes in urban areas and 20 minutes in rural areas. However, in the initial years of the project, the conventional police helpline system was facing numerous challenges such as

limited manpower, lack of infrastructure, operating on obsolete technology, etc. Moreover, in absence of a standardised, coordinated and centralised emergency response system, first responder action was solely dependent upon the ownership of local police and human intervention. Later on, the Government of Haryana hired the services of the company Ernst & Young as its consultant with a mandate of preparing of Detailed Project Report for a modernised and integrated Emergency Response Support System (ERSS) for Statewide rollout. C-DAC was appointed as Total Service Provider (TSP) with the responsibility of supplying, installing, and ensuring operation and maintenance of IT/Non-IT infrastructure, along with the implementation of core solutions and deployment of Technical and operational Manpower.



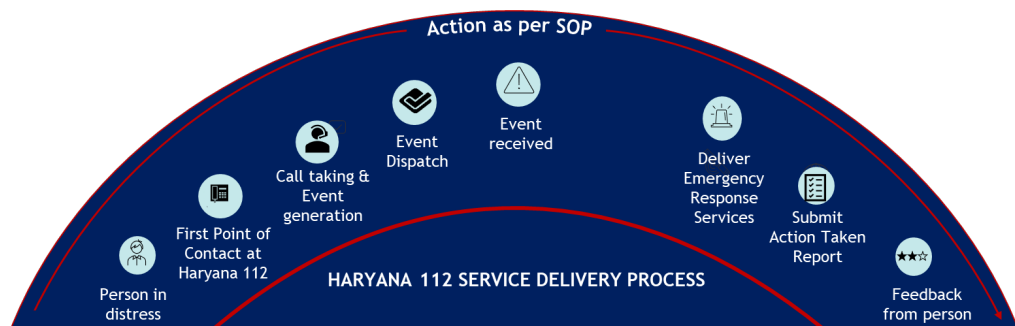
The project was launched on 12.07.2021 by the Hon'ble Chief Minister of Haryana in the gracious presence of the Hon'ble Speaker and Hon'ble Home Minister of Haryana. It completed its First Year of Operation on 12-07-2022. The project boasts an infrastructure which consists of the following core competencies:



- A State-of-Art building, annotated as (SERC), has been designed and constructed at Sector 3, Panchkula. This building has been strengthened with advanced infrastructure to cater to all the emergency services centrally.
- To cater to the additional call load at SERC and to handle any type of technical failure of SERC, a “Mirror Emergency Response Center (MERC)”, with a 20% capacity has been established at Gurugram.
- Redundant connectivity has been built into the system at various levels including DR, Secondary Emergency Response Centre (SERC), connectivity from different ISP, PRI from different telecom providers etc. These redundancies help in overcoming any single point of failure in the system.
- Disaster Recovery Centre (DRC) has been established in the different seismic zone at Hyderabad which holds infrastructure as a backup to manage failover in case of an outage.
- Haryana 112 system has been designed with principles like scalability, open for integration with other systems, robust IT infrastructure, high availability and connecting remote locations.

- 24x7x365 functional Contact Centre established at SERC and MERC (with 20% capacity) operating on centralised call taking and centralised dispatch model. The first and foremost process of ERSS is call taking which is entirely managed round the clock by 157 outsourced professional female staff. Women empowerment and empathic response were the keys consideration while staffing all the female staff as communication officers in the contact centre. These Communication Officers are proficient in all prevalent languages/dialects i.e., Hindi, English, Haryanvi and Punjabi. Adding on, a dedicated desk has been set up for persons with deaf and dumb disabilities. Further, 58 Dispatch Officers manage to round the clock dispatch of the event to the nearest available Emergency Response Vehicles (ERVs). Both CO and DO floors are equipped with large Video walls displaying an informative dashboard which is helpful to monitor daily operations
- 630 brand new Toyota Innova Crysta vehicles along with trained staff, have been deployed across the state for reaching the incident site and act as 'First Responder'. These ERVs are equipped with 23 In-fleet Components and manned by 3 police personnel in two shifts, available at all times for prompt support to the residents in distress. Additionally, 28 Fire Brigades and 42 Ambulances are also covered in ERSS to manage fire and medical-related emergencies in Faridabad and Gurugram.
- More than 5000 dedicated and trained police manpower has been deployed for the smooth functioning of Haryana 112 operations.
- 58 types of Hardware Components comprised of IT and Non-IT equipment are built as a part of robust and scalable Infrastructure. Additionally, 38 Software modules have been exclusively worked out to streamline and integrate operations, processes and information flow for the smooth functioning of ERSS.
- Standard Operating Procedure (SOP) has been developed and implemented in ERSS with the intent to provide operational guidelines/instructions at each step of emergency response. More than 50 types and 170 subtypes of incidents are included in the SOP for the appropriate action based on the severity of the case.
- An in-house training facility has been developed in ERSS to build the capacity of all the Haryana 112 staff to play their respective roles effectively and efficiently. This training facility covers technical, functional and soft skill aspects of capacity building. So far, more than 6000 manpower has been imparted training.

- A rugged Mobile Data Terminal (MDT) as a key communication device is mounted on ERVs to transmit data to and from SERC/MERC. MDT sends its location every 30 seconds of receiving incident data posted from DO.
- The entire process of ERSS is well defined, automated and follows the best practices across the globe. The caller's location is automatically displayed via Location-based-services (LBS), saving valuable time. Events thus created are sent to centralised dispatch. They look up for nearest ERVs on digital maps, assign events, track and follow it up till resolution, and till then, keep on disseminating information to police and city officials concerned. ERVs are required to follow Standard Operating Procedures (SOPs) to ensure proper action is taken, which is then recorded and reported digitally by them. Currently, in Haryana 112, 112/100 (Police Helpline), 1073 (Traffic Helpline), 1930 (Cyber Crime Helpline) and 1091 (Women Helpline) services are running in an integrated mode across the State seamlessly while 108 (Medical) & 102 (Fire) in Faridabad & Gurugram districts only.



Through one call the citizen gets one or multiple services, as required. This convergence reduces citizen distress and enhances response efficiency. Integration with Public Transport Systems, Disaster Management Systems, Indian Railway Helpline Systems and CCTNS is also in progress. 600 two-wheelers are also being added to further reduce response time, which currently stands at a healthy 12.41 minutes statewide (Starting from the time that the caller calls Haryana 112).

The project answers a deeply felt need of all citizens by providing guaranteed emergency services, regardless of the time of the day or the location or background of the caller. Some of the innovations, which have enabled Haryana 112 to bring a paradigm shift in ensuring public safety and building a sense of security in citizens are:

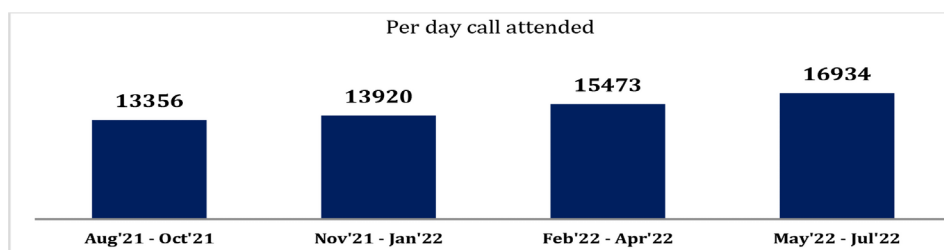
- Enriched GIS map with relevant Points of Interest (Pols) so that precise location could be tracked.
- Automatic call distribution system to ensure no call is left unanswered.
- No territorial boundary limitation within the State for responding to any Emergency.
- 100% follow-up/feedback from citizens to improve services based on suggestions.
- All the patrolling routes have been defined in the System digitally through the MDTs installed in each ERV to ensure patrolling round the clock as per defined routes.
- Consciously choosing non-police officers to answer calls and to take feedback.
- Location Based Services to instantly retrieve caller location and rush help.
- Merit-based monthly reward and recognition system for COs, DOs and ERV staff for motivation.
- Recording and storing all calls, all digital documents and trails to ensure total transparency and accountability.
- Improving evidence collection, to strengthen officers' performance and accountability. Body-worn cameras and Dash Camera are installed in all the ERVs.
- Police ERVs, equipped with 23 In-fleet components (like Stretcher, MDT, First-Aid-Box etc.) provide instant support to the person in distress.
- Four redundant communication networks have been established including GPS signal (for ERV location tracking), MDT, Smart Phone and DMR Sets to ensure definite connectivity with ERVs.

In its first year, the project has had numerous achievements. The major highlights of Haryana 112's operations in the first year are as follows:

Significant increase in calls attended by Haryana 112.

The growing number of calls attended per day by Haryana 112 represents the rise in awareness and trust of citizens towards Haryana 112.

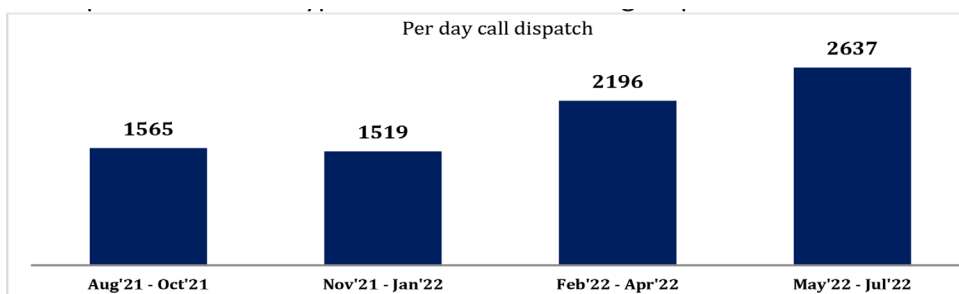
Haryana 112:Emergency Response is just a call away!



Integration with other helplines also contributed to the increase in call volume. Thus, giving a suitable and comfortable platform to the citizens to reach out for help.

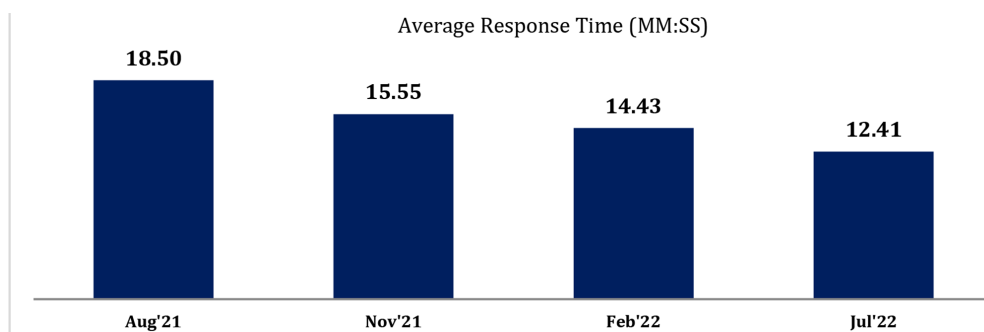
Significant increase in calls dispatched by Haryana 112

Similarly, a significant increase in the volume of call dispatch was seen in the same ratio of calls attended. A growing number of calls dispatched by Haryana 112 represents that all types of incidents are being responded to.



Phenomenal improvement in response time

At the time of Go-live, response time was significantly high due to initial technical glitches and hiccups in the adoption of the new technology/procedures which was remarkably improved in successive months with numerous efforts by the Haryana 112 team such as fixing technical issues and rigorous monitoring of the resources.



High rating of citizen feedback

The level of citizen satisfaction reached up to 90% in the first month of Go-live which shows that users of Haryana 112 services are highly satisfied with the quality of the services delivered. The promptness, empathy towards citizens in times of distress and on-the-spot resolution of most of the cases were found to be the key reasons for the satisfaction of citizens.

The major type of incidents

Quarrelling followed by Accident, Hurt, Crime against women and disputes is the top five types of incidents reported at Haryana 112.

Multiple Success Stories

There are multiple success stories of the exceptional response by ERV staff which was well recognised and appreciated by the community. These cases comprise saving multiple precious lives, foiled robbery, protecting women against crime, rescuing missing persons, etc.

In this way, Haryana 112 has been playing a critical role in responding to emergencies across the State round the clock, assuring the common man that in case of any trouble at any time of the night or day, the police is only a few minutes away.



NCRB ACTIVITIES AND ACHIEVEMENTS

1. ICJS awards 2021



2. Releasing of compendium during the conference on “Good Practices in CCTNS/ ICJS”-2021



3. Winner of Khan Bahadur Azizul Haque Trophy for topping in All India Board Examination for Finger Print Experts



4. National Conference for State Cyber Nodal Officers organised by NCRB (October 2021)



NCRB ACTIVITIES AND ACHIEVEMENTS

5. Stress Management Program organized by NCRB (November 2021)



6. Shri Vivek Gogia, IPS took over the charge of Director on 31/12/2021



7. Vigilance Awareness program organised by NCRB



Note for Future Contributors

The NCRB Journal is intended to encourage practitioners, academia and aspiring writers to submit their original and previously unpublished work in English /Hindi for publication. Articles ranging from short vignettes to fully-developed articles written on reader-friendly topics related to Police Science, Law, Scientific Investigation, Criminology, Forensic Science, Finger Print Science, Biometric Science, Contemporary Legal issues, Cyber-crime, Cyber terrorism, Cyber security, Socio-economic crime, Prisons and Analytical study may be sent. The submission of any solid research or practical methodology that would speak about the needs of those in the police service is also solicited. The journal also encourages articles of serious nature supported by references.

The manuscript should be typed in 12-point, double space with at least a 1-inch margin in MS Word. The manuscript normally should not exceed 5,000 words. However, lengthy articles may also be considered for publication at the discretion of the editorial board. In articles of scientific nature and research, the manuscript should include an abstract of approximately 150 words and 4-5 keywords. All photos and illustrations should be of high quality with good contrast and sharpness. Electronic images are required and must be saved as separate files in JPG, PNG or TIF format with 300 dpi or higher resolution and a minimum of four inches wide. Bar charts, diagrams, sketches etc., should be drawn accurately and clearly. If a chart has supporting data, it should be in Excel format. A caption should accompany each graphic and should accurately describe what is represented in the graphic. For manuscripts in Hindi, the use of Unicode font (Mangal) is suggested. The Manuscripts are subject to both review and editing. All manuscripts accepted for publication shall be the property of NCRB Journal. No article or part of it shall be reproduced or used by the author without the permission of NCRB. Please provide a brief biographical sketch for the author, describing his/her training and expertise in the subject area including the correct name, title, affiliation and contact information with phone number and email address.

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